

Digital adaptation kit for birth defects surveillance

**Operational requirements for implementing
WHO recommendations in digital systems**



World Health
Organization



Digital adaptation kit for birth defects surveillance

**Operational requirements for implementing
WHO recommendations in digital systems**

Digital adaptation kit for birth defects surveillance: operational requirements for implementing WHO recommendations in digital systems
(SMART Guidelines collection)

ISBN 978-92-4-011025-0 (electronic version)

ISBN 978-92-4-011026-7 (print version)

© World Health Organization 2025. Some rights reserved.

This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO licence (CC-BY-NC-SA-3.0-IGO <https://creativecommons.org/licenses/by-nc-sa/3.0/igo/>).

Under the terms of this licence, you may copy, redistribute and adapt the work for non-commercial purposes, provided the work is appropriately cited, as indicated below. In any use of this work, there should be no suggestion that WHO endorses any specific organization, products or services. The use of the WHO logo is not permitted. If you adapt the work, then you must license your work under the same or equivalent Creative Commons licence.

Suggested citation. Digital adaptation kit for birth defects surveillance: operational requirements for implementing WHO recommendations in digital systems. Geneva: World Health Organization; 2025 (SMART Guidelines collection). Licence: [CC BY-NC-SA 3.0 IGO](https://creativecommons.org/licenses/by-nc-sa/3.0/igo/).

If you create a translation or adaptation of this work, you should add the following disclaimer along with the suggested citation: “This translation/adaptation was not created by the World Health Organization (WHO). WHO is not responsible for the content or accuracy of this translation/adaptation. The original English edition shall be the binding and authentic edition”.

If you would like to develop your own Digital Adaptation Kit or create an implementation guide based on this DAK, you can access the SMART Guidelines templates here <https://smart.who.int/ig-starter-kit/>.

Any mediation relating to disputes arising under the licence shall be conducted in accordance with the mediation rules of the World Intellectual Property Organization <https://www.wipo.int/amc/en/mediation/rules>

To submit requests for commercial use and queries on rights and licensing, see <https://www.who.int/copyright>

Third-party materials. If you wish to reuse material from this work that is attributed to a third party, such as tables, figures or images, it is your responsibility to determine whether permission is needed for that reuse and to obtain permission from the copyright holder. The risk of claims resulting from infringement of any third-party-owned component in the work rests solely with the user.

General disclaimers. The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of WHO concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

The mention of specific companies or of certain manufacturers' products does not imply that they are endorsed or recommended by WHO in preference to others of a similar nature that are not mentioned. Errors and omissions excepted, the names of proprietary products are distinguished by initial capital letters.

All reasonable precautions have been taken by WHO to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall WHO be liable for damages arising from its use.

Editing and design by Inis Communication

Contents

Acknowledgements	.iv
Abbreviations	v
Glossary	.vi

Part 1. Overview of SMART guidelines: digital adaptation kits

Background	2
Digital Adaptation Kits within a strategic vision for SMART guidelines.	5
Objectives	7
Components of a DAK.	7
How this DAK was developed	11
Assumptions	12
How to use this DAK.	13

Part 2. Digital adaptation kit content for birth defects surveillance

Component 1. Health interventions and recommendations	22
Component 2. Generic personas	25
Component 3. User scenarios	29
Component 4. Generic business processes and workflows	35
Component 5. Core data elements	52
Component 6. Decision-support logic	57
Component 7. Scheduling logic	58
Component 8. Indicators and performance metrics	59
Component 9. High-level functional and non-functional requirements	61
References	64

Implementation tools

Core data dictionary	https://smart.who.int/dak-bds/dictionary.html
Indicators and performance metrics	https://smart.who.int/dak-bds/indicators.html
Functional and non-functional requirements	https://smart.who.int/dak-bds/system-requirements.html

Acknowledgements

The World Health Organization (WHO) acknowledges the contributions of many individuals and organizations to the development of this digital adaptation kit.

The collation of inputs was coordinated by Theresa Diaz and Gerard Lopez of the WHO Department of Maternal, Newborn, Child and Adolescent Health and Ageing.

Constantin Corman, Akshita Palliwal and Natschja Ratanaprayul, under the leadership of Garrett Mehl of the WHO Department of Digital Health and Innovation, ensured the drafting and content of the digital adaptation kit adhered to WHO standards.

Eugene Wong (United States Centers for Disease Control and Prevention, United States of America, November 2023–January 2025), Salimah R Walani (consultant, WHO Department of Maternal, Newborn, Child and Adolescent Health and Ageing, Switzerland) and Youcef Dahmane (NTT DATA, Belgium) made significant contributions to development of the document.

WHO is grateful to the following people for their input, review and feedback throughout the various stage of development of the Digital Adaptation Kit for Birth Defects Surveillance: Lorenzo Botto (International Clearinghouse for Birth Defects Surveillance and Research, United States of America); Helen Dolk (European Surveillance of Congenital Anomalies, School of Medicine, Ulster University, United Kingdom of Great Britain and Northern Ireland); Pablo Duran (Latin American Center for Perinatology/Women's Health and Reproductive Health, Pan American Health Organization, Uruguay); Boris Groisman (International Clearinghouse for Birth Defects Surveillance and Research, Argentina); Shuchita Gupta (WHO Department of Maternal, Newborn, Child and Adolescent Health and Ageing, Switzerland); Dhiraj Kumar (WHO Department of Information Management and Technology, Switzerland); Diane Lavies (Centre for Infectious Disease Epidemiology and Research, University of Cape Town, South Africa); Cara T Mai (National Center on Birth Defects and Developmental Disabilities, United States Centers for Disease Control and Prevention, United States, September 2021–January 2025); Filippa Pretty (independent consultant, Australia); Jennifer Williams (National Center on Birth Defects and Developmental Disabilities, United States Centers for Disease Control and Prevention, United States, September 2021–January 2025).

Experts from United States-based institutions who made contributions to this publication completed their work prior to 17 January 2025.

This work was funded by the United States Centers for Disease Control and Prevention National Center on Birth Defects and Developmental Disabilities as part of the cooperative agreement 5 E11 DP002196, Global Prevention of Noncommunicable Disease and Promotion of Health (September 2021–September 2023).

Individual, external contributors involved in the development and review of this DAK declared any potential conflicts of interest prior to their involvement. These declarations were reviewed and it was found none of the contributors have conflicting interests that would influence the content of this publication.

Abbreviations

BDS	birth defects surveillance	ICHI	International Classification of Health Interventions
BPMN	business process model and notation	ID	identification
DAK	digital adaptation kit	ISCO	International Standard Classification of Occupations
FHIR	Fast Healthcare Interoperability Resources	LOINC	Logical Observation Identifier Names and Codes
GBDDC	Global Birth Defects Description and Coding	PCPOSS	person-centred point-of-service system
HL7	Health Level Seven International	SMART	standards-based, machine-readable, adaptive, requirements-based and testable
HMIS	health management information system	SNOMED	Systematized Nomenclature of Medicine
ICD	International Classification of Diseases	WHO	World Health Organization
ICF	International Classification of Functioning, Disability and Health		

Glossary

Business process	A set of related activities or tasks performed together to achieve the objectives of the health programme area, such as registration, counselling or referrals (1,2).
Data dictionary	A centralized repository of information about the data elements that contains their definition, relationships, origin, use and type of data. For this DAK, the data dictionary is provided as a spreadsheet.
Data element	A unit of data with a specific and precise meaning.
Decision support (for health workers)	Digitalized job aids that combine an individual's health information with health workers' knowledge and clinical protocols to assist health workers in making diagnoses and treatment decisions (3,4).
Decision-support logic	A set of decision rules for standard and exceptional cases that is separate from the business process. This reduces the complexity of the business process depiction without losing the detail necessary for coding the rules required for system functionality.
Decision-support table	A semistructured way to depict each discrete decision that will need to be embedded in the system. Depending on the complexity of the clinical guidelines, there will likely be multiple decision-support tables.
Digital health	The systematic application of information and communication technologies, computer science and data to support informed decision-making by individuals, the health workforce and health systems, to strengthen resilience to disease and improve health and wellness (1,5).
Digital tracking	The use of a digitalized record to capture and store health information to enable follow-up of clients' health status and services received. This may include digital forms of paper-based registers and case management logs within specific target populations, and electronic medical records linked to uniquely identified individuals (3,4).
Functional requirements	The capabilities a system must have to meet end-user needs and achieve tasks within the business process.
Health information system	A system that integrates data collection, processing, reporting and use of the information necessary to improve health service effectiveness and efficiency through better management at all levels of the health service (6).
Health management information system (HMIS)	An information system specifically designed to assist in the management and planning of health programmes rather than delivery of care (6).
Interoperability	The ability of different applications to access, exchange, integrate and use data in a coordinated manner through the use of shared application interfaces and standards, within and across organizational, regional and national boundaries, to provide timely and seamless portability of information and optimize health outcomes.
Non-functional requirement	The general attributes and features of a digital system to ensure usability and overcome technical and physical constraints. Examples of non-functional requirements include the ability to work offline, multiple language settings and password protection.
Persona	A generic aggregate description of a person involved in or benefiting from a health programme.
Person-centred point of service system (PCPOSS)	A digital system that facilitates the provision of health services to individuals at the point of service or point of care. This includes software capabilities that enable health workers to access, record and update health information, and software capabilities that enable health workers to communicate with clients. PCPOSS encompass various services and application types, including community-based information systems, decision-support systems, electronic medical or health record systems and personal health records (3,7).
Standard	In software, a specification used in digital application development that has been established, approved and published by an authoritative organization. These rules allow information to be shared and processed in a uniform, consistent manner independent of a particular application.
Task	A specific action in a business process.
Terminologies	In clinical care, structured vocabularies covering health-related concepts, such as diseases, diagnoses, laboratory tests and treatments, to enable the storage, analysis and exchange of data in a consistent and standard way (8).
Workflow	A visual representation of the progression of activities (tasks, events, decision points) in a logical flow illustrating the interactions within the business process (2).

Part 1.

Overview of SMART guidelines: digital adaptation kits



Background

As digital technologies are increasingly being leveraged to enable and support health service delivery and accountability, health ministries and partners have recognized the value of digital health as articulated within the World Health Assembly Resolution on digital health (9) and the World Health Organization (WHO) Global Strategy on Digital Health 2020–2025 (10). Funding agencies have advocated for the rational use of digital tools as part of efforts to expand coverage and quality of services, promote data use and monitor efforts (11–13).

Guidelines are often available only in a narrative format that requires a resource-intensive process to be elaborated into the specifications needed for operationalizing digital systems. This translation of guidelines for digital systems often results in subjective interpretation for implementers and software vendors, which can lead to inconsistencies or inability to verify the content within these systems, potentially leading to adverse health outcomes and other unintended effects (14–17).

Despite the abundance of digital tools developed and deployed for health, there is often limited transparency in the data model and logic model, and the evidence-based clinical or public health recommendations contained within these digital tools, because system documentation may be unavailable or proprietary, requiring governments to start from scratch and expend additional resources each time they intend to update and deploy such a system.

This lack of health content documentation undermines the credibility of such systems, leading to dependency on one vendor and haphazard deployments that are unscalable, impeding the opportunity for interoperability that would otherwise enable continuity of care (17–19).

WHO standards-based, machine-readable, adaptive, requirements-based and testable (SMART) guidelines provide essential ingredients to facilitate digital health transformation of health programmes in a way that is consistent with recommended clinical, public health, data practices and interoperability standards. To ensure countries can effectively benefit from digital health investments, digital adaptation kits (DAKs), which are the second operational guidelines layer of the SMART guidelines approach, are designed to facilitate the accurate reflection of WHO clinical, public health and data use guidelines within the digital systems that countries are adopting. DAKs are operational, software-neutral, standardized documentation that distil clinical, public health and data use guidance into a format that can be transparently incorporated into digital systems (19).

Although digital implementations comprise multiple factors – including the health domain data and content, the digital intervention or functionality, and the digital application or communication channel for delivering the digital intervention – DAKs focus primarily on ensuring the validity of the health content (Fig. 1) (1,4). DAKs provide the generic content requirements that should be housed within digital systems, independent of a specific software application and with the intention that countries can customize them to local needs.

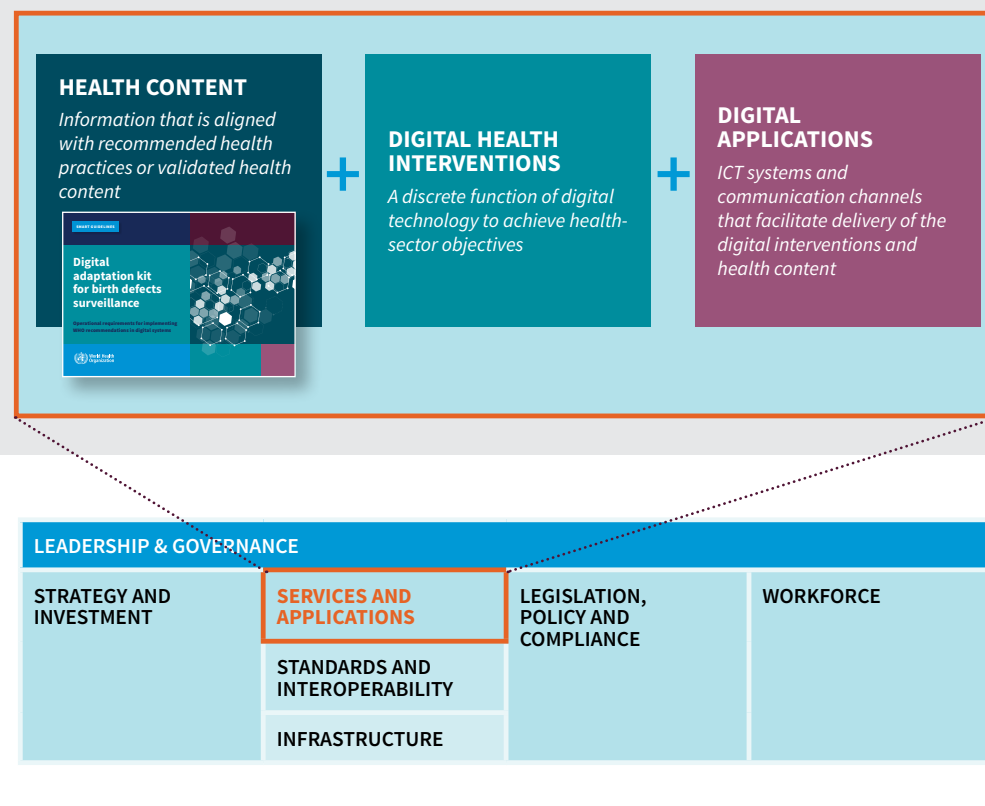
For this particular DAK, the requirements are based on systems that provide the functionalities of person-centred point-of-service systems (PCPOSS) (Box 1) and include components such as personas, workflows, core data elements and reporting indicators. Operational outputs, such as spreadsheets for the data dictionary and the detailed calculation of indicators, are included as part of the DAK as practical resources that implementers can use as starting points when developing digital systems. Data components within the DAK are mapped to standards-based classifications and terminologies, such as the International Classification of Diseases (ICD), International Classification of Functioning, Disability and Health (ICF), International Classification of Health Interventions (ICHI), Logical Observation Identifier Names and Codes (LOINC), and Systematized Nomenclature of Medicine (SNOMED), to facilitate interoperability.

Box 2 provides context-specific recommendations for the use of an integrated PCPOSS.

The DAKs follow a modular approach in detailing the data and content requirements for a specific health programme area – such as tuberculosis, family planning or antenatal care – among the different health areas for which DAKs have been developed. **This DAK focuses on providing the content requirements for PCPOSS used in health-care facilities for birth defects surveillance.**

Fig. 1

Digital adaptation kits and their role in digital health implementations



ICT: information and communications technology.

Source: Adapted from (4)

Box 1

What is a person-centred point-of-service system?

A **person-centred point-of-service system (PCPOSS)** is digital in nature and facilitates the provision and delivery of health services to individuals (i.e. persons, clients, patients, health service users) at the point of care. A PCPOSS includes software capabilities that enable health-care providers to access, record and update individuals' health information and to interactively communicate with individuals. A PCPOSS encompasses various services and application types (3,7):

- **Community-based information systems** “facilitate data collection and use at the community level. These applications are utilized by community-based workers who provide health promotion and disease prevention activities”.
- **Decision support systems** are digital “tools that combine medical information databases and algorithms with patient specific data. They are intended to provide health-care professionals and/or users with recommendations for diagnosis, prognosis, monitoring and treatment of individual patients”.
- **Electronic health record systems** are “secure, online system[s] that hold information about people's health and clinical care and [are] managed by health workers”.
- **Personal health records** are “record[s] of an individual's health information in a structured digital format for a set of defined use cases over which the person has agency”.

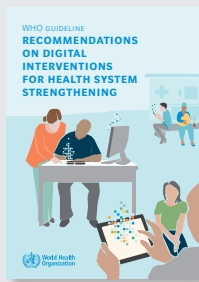
PCPOSS end-users may include all health worker occupational groups operating at all care levels, including those operating outside formal health-care facilities such as community health workers and health volunteers.

Note that this DAK for birth defects surveillance does not include decision support because it is not programmatically relevant in this context.

Box 2

WHO recommendation related to person-centred point-of-service system

WHO has provided the following context-specific recommendation for the use of an integrated system that provides both digital tracking of clients' health status and services and decision support (4):



Effective coverage Accountability coverage

Digital tracking of clients' health status and services (digital tracking) combined with decision support

RECOMMENDATION 8: WHO recommends digital tracking of clients' health status and services, combined with decision support under these conditions:

- ▶ in settings where the health system can support the implementation of these intervention components in an integrated manner; and
- ▶ for tasks that are already defined as within the scope of practice for the health worker.

(Recommended only in specific contexts or conditions)

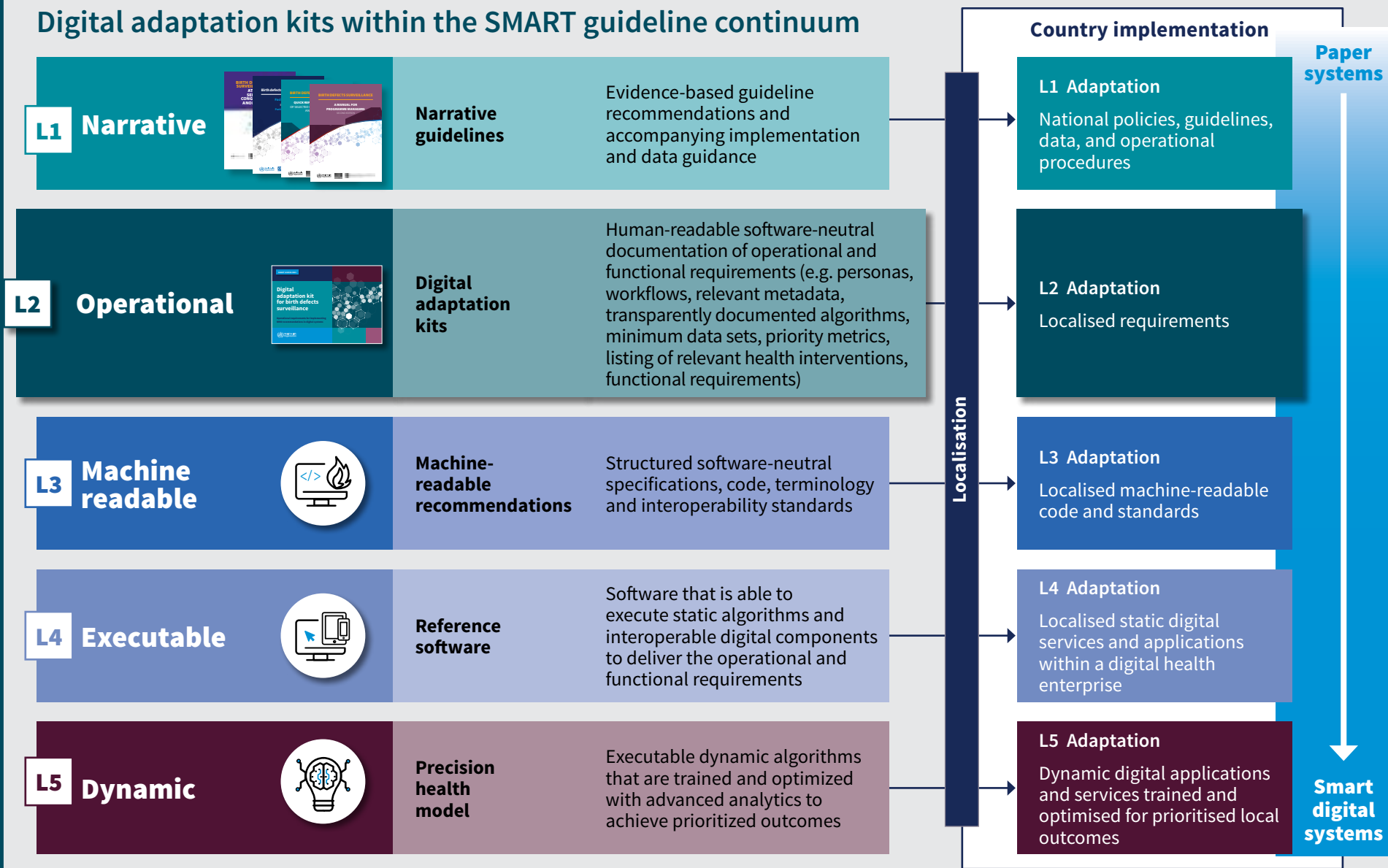
Based on the principle of “collect once, use many times” (20), this DAK outlines the requirements of a PCPOSS that facilitates the use of more reliable source data for aggregate indicators at management levels by providing access to primary data collected directly at the point of care. This allows a shift away from the need for aggregate indicators to be reported separately and for investments in paper-based or digital indicator reporting systems. Data collected during service delivery can also be used for secondary data reporting for accountability, including monitoring provider, stock and system performance. Factors affecting overall health system performance can be highlighted more promptly and accurately, and the clerical burden on health workers is reduced.

Digital Adaptation Kits within a strategic vision for SMART guidelines

The operational and standardized documentation reflected within the DAKs represents one of the steps within a broader vision of SMART guidelines. SMART guidelines aim to maximize health impact through improved fidelity and uptake of recommendations within standards-based digital systems via a systematic process for transforming guideline development, delivery and application (16,21). Within this vision, DAKs serve as a prerequisite for developing computable or machine-readable guidelines, and as executable reference software and advanced analytics for precision health. Fig. 2 provides an overview of the different layers of the SMART guidelines continuum and where DAKs fit within this strategy (16,21,22).

Fig. 2

Digital adaptation kits within the SMART guideline continuum



Source: Adapted from (21).

Objectives

This DAK focuses on birth defects surveillance (BDS) and aims to provide a common language across various audiences – including birth defects surveillance programme managers, software developers and implementers of digital systems – to ensure a common understanding of the appropriate health information content requirements within a defined health programme area, as a mechanism to catalyse the effective use of these digital systems. The key objectives of the DAK are to:

- ensure adherence to WHO clinical, public health and data use guidelines, and facilitate consistency of the health content used to inform the development of a PCPOSS;
- enable health programme leads and digital health teams (including software developers) to have a joint understanding of the health content within the digital system, with a transparent mechanism to review the validity and accuracy of the health content;
- provide a starting point of the core data elements and indicator calculation that should be included within a PCPOSS for birth defects surveillance.

Information detailed in this DAK reflects generic workflow processes and data, as derived from the WHO Birth defects surveillance: A manual for programme managers (23) and other related WHO documents. The outputs of the DAKs are intentionally generic and will need to be contextualized to local policies and requirements.

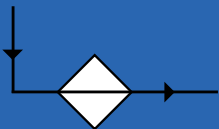
DAKs have also been developed for antenatal care, family planning, tuberculosis, child health in humanitarian settings, immunizations and HIV. The approach is being expanded to additional health domains, such as postnatal care and community health. Many of the other DAKs have been developed for point-of-care data collection in primary care settings, where operational requirements regarding care are the main purpose, with data for monitoring and surveillance being an integrated component. This DAK differs in that its main purpose is surveillance, and its data may be drawn from and feed into existing clinical care systems. All the DAKs work towards a comprehensive approach for standardized software requirements for primary health-care settings.

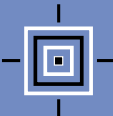
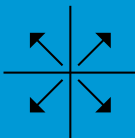
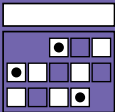
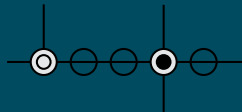
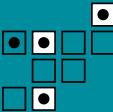
Components of a DAK

The WHO SMART guidelines for a DAK comprise nine interlinked components: health interventions and associated recommendations; generic personas; user scenarios; generic business processes and workflows; core data elements; decision-support logic; scheduling logic; indicators and reporting requirements; and high-level functional and non-functional requirements. Table 1 overviews the contributing components of a DAK. All information within the DAK represents a generic starting point, which can be adapted according to the specific context.

This DAK does not include the decision-support logic and scheduling logic components because it does not cover recommended clinical practice related to birth defects.

Table 1 Components of a DAK

Component	Description	Purpose	Outputs/artifacts	Adaptation needed
1. Health interventions and recommendations 	Overview of health interventions and WHO recommendations included within this DAK. DAKs are meant to be a repackaging and integration of WHO guidelines and guidance documents in a particular health domain. The list of health interventions is drawn from the universal health coverage menu of interventions compiled by WHO (24,25).	Setting the stage to understand how this DAK would be applied to PCPOSS in the context of specific health programmes and interventions.	» List of related health interventions based on WHO universal health coverage essential interventions. » List of related WHO recommendations based on guidelines and guidance documents.	» Contextualization to reflect current or planned national policies.
2. Generic personas 	Depiction of end-users and related stakeholders who would be interacting with the digital system or involved in the care pathway.	Contextualization to understand the wants, needs and constraints of end-users.	» Description, competencies and essential interventions performed by key personas.	» Greater specification and details on end-users based on real people (e.g. health workers) in a given context. » High-level information to describe the provider of the health service (e.g. the general background, roles and responsibilities, motivations, challenges, and environmental factors).
3. User scenarios 	Narratives that describe how different personas may interact with the digital system and with each other. User scenarios are illustrative only and are intended to give an idea of a typical workflow.	Contextualization to understand how the system would be used, and how it would fit into existing workflows.	» Example narrative of how key personas may interact with the system and with each other during a workflow.	» Greater specification and details on the real needs of end-users in a given context.
4. Generic business processes and workflows 	A business process is a set of related activities or tasks performed together to achieve the objectives of the health programme area, such as registration, counselling or referrals (1,2). A workflow is a visual representation of the progression of activities (tasks, decision points, interactions) performed within the business process (1,2).	Contextualization and system design to understand how the digital system would fit into existing workflows and how best to design the system for that purpose.	» Overview matrix presenting key processes in birth defects surveillance. » Workflows for identified business processes with annotations. » Link to business process workflows (https://smart.who.int/dak-bds/business-processes.html).	» Customization of workflows, which may include additional forks, alternative pathways or entirely new workflows.

Component	Description	Purpose	Outputs	Adaptation needed
5. Core data elements 	Data elements are required throughout the different points of the workflow. These data elements are mapped to standards-based classifications and terminologies to ensure the data dictionary is compatible with other digital systems.	System design and interoperability to know which data elements need to be logged and how they map to other standard terminologies (e.g. ICD and LOINC) for interoperability with other standards-based systems.	» List of data elements. » Link to data dictionary (https://smart.who.int/dak-bds/dictionary.html) with detailed data specifications in spreadsheet format.	» Translation of data labels into local languages and additional data elements created, depending on context.
6. Decision-support logic 	Decision-support logic and algorithms to support appropriate service delivery in accordance with WHO clinical, public health and data use guidelines.	System design and adherence to recommended clinical practice to know what underlying logic needs to be coded into the system.	» List of decisions that need to be made throughout the encounter. » Link to decision-support tables in a spreadsheet format with inputs, outputs and triggers for each decision-support logic. (Note: this component is not relevant for the context of birth defects surveillance.)	» Change of specific thresholds or triggers in a logic (IF/THEN) statement, such as body mass index cut-off or age trigger for youth-friendly services. » Additional decision-support logic formulas, depending on context.
7. Scheduling logic 	Scheduling logic to support appropriate reminders for follow-up visits and services in accordance with WHO clinical, public health and data use guidelines.	System design to know what logic needs to be coded into the system so that appropriate reminders are generated.	» List of scheduling logic tables. » Link to scheduling logic for services in spreadsheet format. (Note: this component is not relevant for the context of birth defects surveillance.)	» Additional scheduling logic, depending on context.
8. Indicators and performance metrics 	Core set of indicators that need to be aggregated for decision-making, performance metrics, and subnational and national reporting. These indicators and metrics are based on data that can feasibly be captured from a routine digital system rather than survey-based tools.	System design and adherence to recommended health monitoring practices to know what calculations and secondary data use are needed for the system, based on the principle of “collect once, use many” (20).	» Link to indicators table (https://smart.who.int/dak-bds/indicators.html) with numerator and denominator of data elements for calculation, along with appropriate disaggregation.	» Changing calculation formulas of indicators. » Adding indicators. » Changing the definition of the primary data elements used to calculate the indicator based on available data.
9. High-level functional and non-functional requirements 	List of core functions and capabilities that the system must have to meet end-users' needs and achieve tasks within the business process.	System design to know what the system should be able to do.	» Link to tables of functional and non-functional requirements (https://smart.who.int/dak-bds/system-requirements.html) with the intended end-user of each requirement, and why that user needs that functionality in the system.	» Adding or reducing functions and system capabilities based on budget and end-user needs and preferences.

Throughout the DAK, identification (ID) numbers are used to simplify tracking and referencing of each of the components. The DAK represents an overview across the different components, and the comprehensive and complete outputs of each component (e.g. data dictionary and indicators tables) are included as linked implementation tools. Box 3 provides an overview of the notation guidance.

Box 3

ID notation guidance

- **Component 1 – health interventions and recommendations:**
 - No notations used.
- **Component 2 – generic personas:**
 - No notations used.
- **Component 3 – user scenarios:**
 - No notations used.
- **Component 4 – generic business processes and workflows:**
 - Each workflow should have a “process name” and a corresponding letter.
 - Each workflow should have a “**process ID**” structured as “**abbreviated health domain**” and “**corresponding letter for the process**” (e.g. BDS.B).
 - Each activity in the workflow should be numbered with an “**activity ID**” structured as “**process ID**” and “**activity number**” (e.g. BDS.B7).
- **Component 5 – core data elements (data dictionary):**
 - Each data element should have a “**data element (DE) ID**” structured as “**activity ID**”, “DE” and “sequential number of the data element” (e.g. BDS.B7.DE.1, BDS.B7.DE.2).
- **Component 6 – decision-support logic:**
 - Not applicable.
- **Component 7 – scheduling logic:**
 - Not applicable.
- **Component 8 – indicators and performance metrics:**
 - Each indicator should have an “**indicator (IND) ID**” structured as “**abbreviated health domain**”, “IND” and the “sequential number of the indicator” (e.g. BDS.IND.1, BDS.IND.2).
- **Component 9 – high-level functional and non-functional requirements:**
 - Each functional requirement should have a “**functional requirement (FXNREQ) ID**” structured as “**abbreviated health domain**”, “FXNREQ” and “sequential number of the functional requirement” (e.g. BDS.FXNREQ.1, BDS.FXNREQ.2).
 - Each non-functional requirement should have a “**non-functional requirement (NFXNREQ) ID**” structured as “**abbreviated health domain**”, “NFXNREQ” and “sequential number of non-functional requirement” (e.g. BDS.NFXNREQ.1, BDS.NFXNREQ.2).

How this DAK was developed

Birth defects are a major cause of childhood deaths and disabilities. The Sixty-third World Health Assembly adopted a birth defects resolution in 2010 (26), which was expanded with Resolution WHA77.5 in June 2024 (27). This Resolution encourages countries to build capacity related to the surveillance, prevention and care of birth defects and to raise awareness about their effects. In an effort to prevent birth defects and to care for people with birth defects, it is important to get a full understanding of the prevalence of these conditions. WHO published the Birth defects surveillance toolkit, which includes the Quick reference handbook (28), Manual for programme managers (23) and Facilitator's guide (29).

In 2020, an international committee that included representatives from WHO developed Global Birth Defects Description and Coding (GBDDC) (30), a mobile application for the description and coding of congenital anomalies that are externally visible for use in low- and middle-income countries aligned with the WHO Birth Defects Surveillance Toolkit. The application is available in Google Play Store (31) and Apple App Store (32). The application has been used in over 40 countries. After review of the application and discussions with key stakeholders, it was agreed that a DAK needed to be developed by WHO so that health programme managers can develop applications for their own settings that fit their digital ecosystems and are maintained directly in countries, in alignment with WHO specifications and recommendations.

To develop the DAK, a mapping of existing WHO guidelines, guidance and tools relevant for the development of a DAK for birth defects surveillance was carried out in addition to documenting the requirements in the GBDDC application. Key resources that were identified included the WHO Birth defects surveillance: A manual for programme managers (23).

A desk review of these resources was conducted. The recommendations from clinical guidelines were extracted and synthesized to form the components of the birth defects surveillance DAK. This process guided the development of the workflows for key birth defects surveillance programmatic and data processes, and a data dictionary. The indicators and performance metrics included in the birth defects surveillance DAK were informed by the Manual for programme managers (23). The data elements required to calculate the indicators were based on the relevant clinical guidelines and guidance documents. The data elements were mapped on to standardized terminologies and classifications by a medical terminologist to facilitate the adoption of interoperability standards into digital systems.

All components of the birth defects surveillance DAK were refined through a series of in-depth technical consultations with the United States Centers for Disease Control and Prevention and relevant teams from the WHO Department of Maternal, Newborn, Child and Adolescent Health and Ageing, which were responsible for developing the WHO guidelines and guidance used to guide the development of this DAK. DAKs are derivative guidance documents to support the implementation of WHO guidelines and guidance in digital systems. These teams reviewed and validated the sections of the birth defects surveillance DAK relevant to their area of expertise to ensure each component reflects the narrative in WHO guidelines and guidance. Other published DAKs, and the teams responsible for their development, were consulted as needed to ensure alignment across the SMART guidelines programme.

Individual, external contributors involved in the development and review of this DAK declared any potential conflicts of interest prior to their involvement. These declarations were reviewed and it was found none of the contributors have conflicting interests that would influence the content of this publication.

Assumptions

The following critical assumptions were made in the development of this DAK:

- For the purposes of this DAK, the definition of a “case” is a unique newborn delivered with at least one of the selected birth defects for inclusion in the surveillance system.
- Local adaptations will be made to this DAK, because this document is intended to reflect commonalities generalized from different settings. The content is assumed to be 80% generic for use across different settings, while 20% of the content will require local contextualization. Local adaptations must be made for the inclusion and exclusion criteria that define case eligibility. This DAK includes data elements for common variables used for inclusion and exclusion criteria and for an initial list of major external and internal birth defects to consider for monitoring.
- Birth defects surveillance requires established systems for the collection and analysis of large amounts of data from various sources, including health-care providers, laboratories and public health agencies. Specifically, it must also have the total number of births within the same surveillance population to accurately calculate prevalence indicators.
- Timely and accurate identification of birth defects is critical to inform public health policies, prevention strategies and clinical care.
- Birth defects may have complex and multifactorial causes, including genetic, environmental and behavioural factors, and their occurrence may vary by region, race and socioeconomic status.
- Birth defects surveillance requires a multidisciplinary approach, involving collaboration among health-care providers, public health agencies, researchers and community stakeholders.
- Birth defects are present at birth and may be detected prenatally, shortly after birth, in infancy or later in childhood. This DAK concentrates on detection of birth defects in the neonatal period (the first four weeks after birth). Further adaptations could be added for a wider age range, including birth defects detected prenatally.
- Continuous monitoring and evaluation of birth defects surveillance systems are necessary to ensure their effectiveness in ascertaining cases, efficiency in reporting and verifying cases, and relevance over time to the goals of end-users.
- Although many countries do not require informed consent for public health surveillance, it is important to ensure the collection and use of data and photographs in the surveillance comply with the national and local regulations and ethical standards in the country where it is being implemented. This may require developing appropriate consent forms, providing clear information to clients and families, and obtaining approval from institutional review boards or other oversight bodies.

- The BDS system may operate as a standalone system or as part of a larger system within the facility or public health context. The BDS system may need to receive data from other systems or provide data to other systems, as needed. In particular, antenatal care, neonatal health and newborn health modules that are not part of the birth defects surveillance DAK may be used to capture and analyse data related to newborn health and outcomes. These modules may include further information on Apgar scores and other clinical and demographic factors that may be relevant to understanding the occurrence and impact of birth defects. The design and implementation of the neonatal module may require additional resources and expertise beyond those needed for the birth defects surveillance DAK.
- The identification and registration of newborns could happen through interoperability with other individual-level record systems, such as an electronic medical record. In these instances, the birth defects surveillance systems could retrieve registration data directly from the electronic medical record and make use of it in the BDS context, as needed – although this is not required.
- Clinical care for birth defects, including referral, is beyond the scope of this DAK.

How to use this DAK

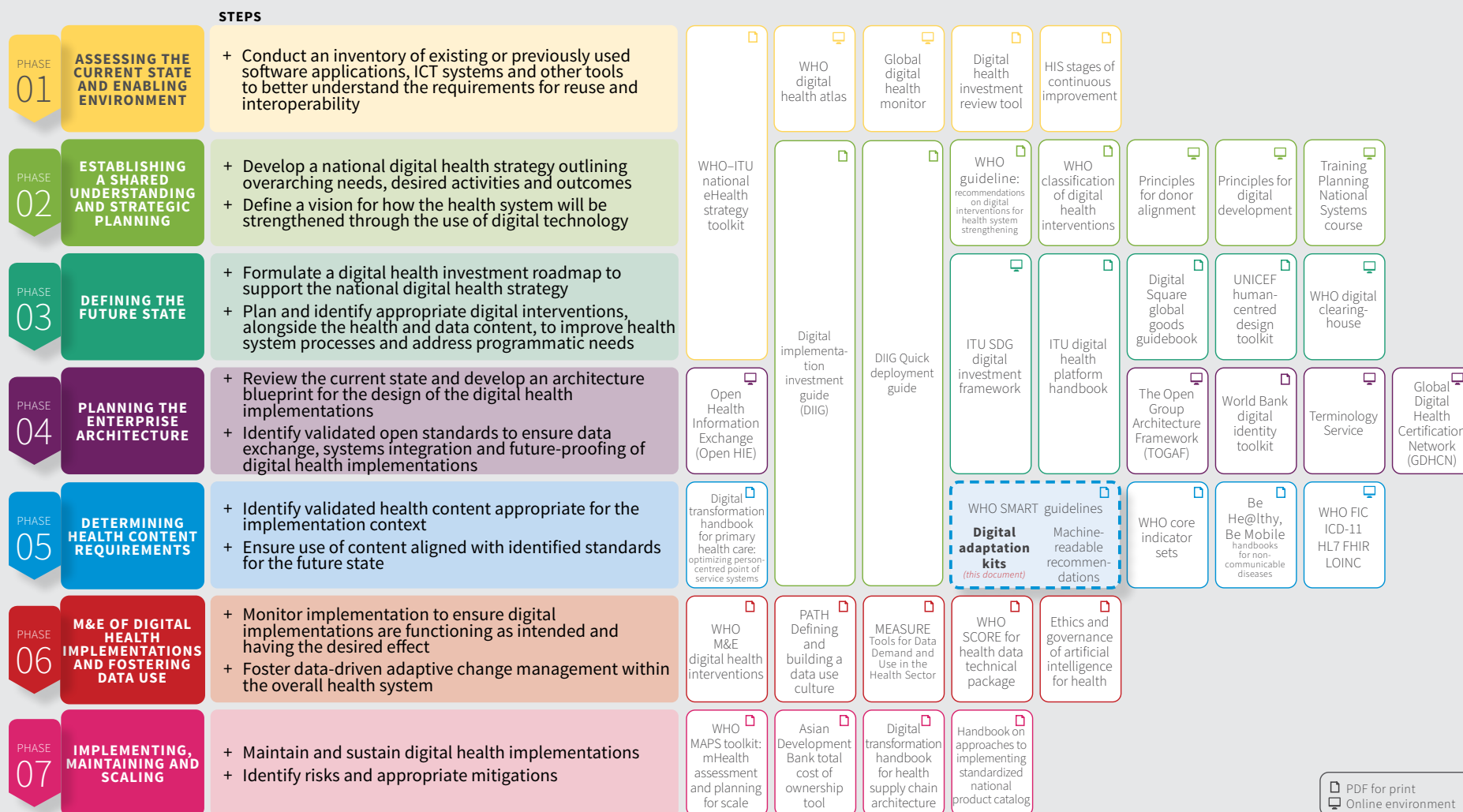
Once the strategic vision for a point-of-service system has been developed, SMART guidelines can assist in determining the necessary health content requirements based on available guidance. In contexts where an overarching vision may not exist, users should first consult the WHO and International Telecommunication Union National eHealth strategy toolkit (33), the WHO Guideline: Recommendations on digital interventions for health system strengthening (4), and the WHO Digital implementation investment guide (DIIG) (1) to establish a better understanding of how to select and apply appropriate digital health interventions. Fig. 3 situates DAKs within the broader set of resources for planning and implementing digital health systems.

Target audience

The primary target audience for this DAK consists of health programme managers within health ministries who will be working with their digital or health information systems counterparts to determine the health content requirements for a birth defects surveillance PCPOSS. The health programme manager is responsible for overseeing and monitoring the implementation of the clinical practices and policies for the health programme area (in this case, birth defects surveillance).

Fig. 3

Planning and implementing a digital health enterprise: phases, steps and resources



PDF for print
 Online environment

HIS: health information system; ICT: information and communications technology; ITU: International Telecommunication Union; M&E: monitoring and evaluation; SDG: Sustainable Development Goal.

Source: Adapted from WHO (1).

The DAK also equips individuals responsible for translating health-system processes and guidance documents for use within digital systems with the necessary components to kickstart the process of developing a PCPOSS in a standards-compliant manner. These individuals are also known as business analysts who interface between health content experts and software development teams. Specifically, the DAK contains key outputs such as the data dictionary to ensure the validity and consistency of the health content within the PCPOSS.

Using this DAK requires collaboration between health programme managers responsible for birth defects surveillance and counterparts in digital health and health information systems. Although each DAK focuses on a particular health programme area (in this case, birth defects surveillance), the DAKs are envisioned to be used in a modular format and to link to other health programme areas within primary health-care settings to support integration across services.

Scenarios for using this DAK

This DAK may be used across various scenarios, some of which are listed below.

Scenario 1

Incorporating WHO guideline content into an existing PCPOSS

Countries that already have digital systems in place, such as electronic medical records, decision-support tools or surveillance tools, may use the information in the DAK to cross-check whether the underlying content and data for specific health programme areas are aligned with WHO guidelines. The DAK can be used to identify and extract specific data that need to be incorporated into existing digital systems. By reviewing this systematic documentation, health programme managers and implementers can more readily identify differences in workflows and data inputs to examine the rationale for deviations and understanding local adaptations of guideline content.

Scenario 2

Transitioning from paper to digital

Some countries may have paper-based systems that they would like to digitalize. The process of optimizing paper-based client-level systems into digital records and decision support may be overwhelming. The DAK can be used as a starting point for streamlining the data elements that should be in the optimized client-level digital system. The paper-based tools can then be checked to determine whether there are any missing fields or content that should be included within the digital system.

The WHO Digital transformation handbook for primary health care (7) provides stepwise guidance on mapping data from paper-based forms into a digital system, including accounting for data elements that are redundant or do not add value to the health system.

Scenario 3 Linking aggregate health management information systems to a PCPOSS used at the point of care	Some countries may already have digital systems for aggregate reporting and health management information systems (HMIS) but may not have implemented digital systems that function at the service-delivery level. The DAK can guide the development of a PCPOSS that includes an electronic medical or health record system, a community-based information system, a decision-support system and personal health records that operate at the point of service or care and ensure linkages between the data collected at service-delivery levels (e.g. community or facility level) and at aggregate levels. A component of the DAK provides aggregate indicators derived from individual-level data to provide the linkage between these different levels (e.g. aggregate reporting at facility, community, district and national levels). Complementary guidance dedicated specifically to aggregate-level data, such as the WHO Analysis and use of health facility data: Guidance for maternal, newborn, child and adolescent health programme managers (34) and the WHO SCORE for health data technical package (35), should be consulted to support the use of routine data at the facility management and district levels.
Scenario 4 Leveraging data standards to promote interoperability and integrated systems	This DAK includes data elements mapped to ICD 10th and 11th revision, ICF, ICHI, LOINC and SNOMED codes to support the design of interoperable systems. The data dictionary provides the codes for different data elements, reducing the time needed to incorporate these global standards into the design of a digital system. Box 4 provides guidance on developing the third layer of the SMART guidelines and its role in designing interoperable systems.
Scenario 5 Linking point-of-service systems used at the point of care by health workers to personal health records systems	A critical part of service delivery in any health domain relies on engaging with clients. Digital interventions aimed at clients – such as on-demand information services, targeted client communication (e.g. transmitting health information and reminders), reporting of health-system feedback by clients on the quality of care, accessing their own medical or home-based records, and self-monitoring their health and diagnostic data (3) – are emerging approaches for complementing the services provided by health workers.
Scenario 6 Updating paper registers	Countries that continue to use paper registers and are not ready to digitalize can use the DAK content to update existing paper registers and decision-support tools by simplifying them, reducing data collection, and updating paper-based tools to align with the latest evidence-based guidelines and WHO recommendations. In this way, nondigital processes can be supported by the DAK.
Scenario 7 Establishing benchmark requirements to facilitate governance of digital health systems	The DAK allows countries to clearly outline software-agnostic baseline expectations and requirements for a digital service and application for a particular use. The templated format allows countries to establish their own standards for any digital service and application that is to be implemented in country. The DAK provides a shared language to allow for alignment across all the relevant stakeholders, regardless of which software system is used.

Box 4

Third layer of the SMART guidelines

The next step in the digitalization journey of normative content involves the development or implementation of the computable and machine-readable content of SMART guidelines – also known as the third knowledge layer (L3). This layer builds on L2 and is represented as a Fast Healthcare Interoperability Resources (FHIR) implementation guide. L3 focuses on producing machine-readable recommendations that can be tested for conformance to interoperability standards. This step ensures the semi-structured normative content, in the form of the DAK, is converted to a machine-readable format that promotes interoperability and can be adapted easily to meet the requirements of different contexts.

Developing L3 provides benefits to multiple stakeholders, including:

- the ministry of health – a localized implementation guide promotes a local ecosystem of standards that many software vendors can tap into, thereby avoiding software lock-in;
- technology partners – with standards-based machine-readable code available within the implementation guide, partners have a quicker development cycle and easier adoption of standards-based code into solutions; and
- health workers – access to decision-support logic within digital systems provides greater support for day-to-day working.

Stakeholders who want to understand the importance of interoperability and its key components and to plan investments in integrating these components in the national digital health systems should consult the Stepwise toolkit for planning and budgeting interoperability of digital health solutions (36).

Before developing or implementing L3, it is important to:

- identify the requirements of the digital tool, such as translation needs and test data generation needs;
- determine the necessary budget and required resources;
- assemble a team with the required software development skills and experience to develop or implement an FHIR implementation guide; and
- review the L3 standard operating procedures, intended to guide developers and implementers through the process of creating a Health Level Seven International (HL7) FHIR implementation guide following the WHO SMART guidelines approach. The SMART guidelines community is actively working on defining the standard operating procedures for authoring L3 implementation guides (37).

The status of all L3 implementation guides developed by WHO and implementation partners is available at <https://smart.who.int/>.

Steps for SMART guidelines implementation

A key measure of success for a DAK is its adoption and use at the country level (21). The localization and implementation of a DAK should be undertaken as part of a wider strategy for SMART guidelines implementation and as part of the wider strategy for digital transformation of health systems. The five high-level steps are outlined below.

Step 1: planning and assessment

Before implementing SMART guidelines, a planning and assessment phase brings together all the key stakeholders involved in digital transformation, either nationally or for a specific programme area. Once there is a clear vision and strategic direction for adoption of SMART guidelines, the SMART guidelines, including the DAKs, become a valuable resource to support a country to define the requirements for a digital health system within the country's broader digital health ecosystem. The use of SMART guidelines should be incorporated into current and planned processes for developing digital services and applications in health.

Step 2: country-level partnerships

Building a coalition of stakeholders, including the ministry of health, relevant government departments, funding agencies, nongovernmental organizations and the private sector, is essential for long-term success. This involves engaging with stakeholder groups to raise awareness and advocate for the adoption of SMART guidelines. Once sufficient commitment is secured, a national implementation roadmap should be developed with clear objectives, timelines, responsibilities and resource needs.

Step 3: localization

The WHO SMART guidelines, including the DAKs, help provide a benchmark, but localization of the SMART guidelines, including the DAKs, is required to establish a shared language in which countries can adapt and incorporate into digital health governance processes. Localizing SMART guidelines to the national context involves conducting multistakeholder requirements-gathering workshops for each health programme. These workshops aim to localize DAKs and HL7 FHIR implementation guides so they align content with national policies, protocols and terminology. The outcomes of these workshops are localized versions of DAKs and HL7 FHIR implementation guides tailored to the national context and translated into local languages. These localized SMART guidelines and localized DAKs are what would be incorporated into a digital system for implementation.

Step 4: technical implementation

The choice of technical implementation model for SMART guidelines will vary from country to country. Depending on the national digital health strategy and national digital public infrastructure, existing digital health solutions may be enhanced or refactored to align with SMART guidelines, or new tools may need to be developed altogether to ensure adherence to SMART guideline requirements and interoperability standards.

At this stage, change-management workshops will need to be organized for key stakeholders, including developers, health programme managers and frontline health workers at the country level, on the use of digital services and applications that have been built or updated to execute SMART guideline content. Rigorous quality assurance and user acceptance testing will need to take place before any wider rollout to identify any bugs and ensure fidelity with evidence-based guidelines. The WHO Digital implementation investment guide (DIIG): Integrating digital interventions into health programmes (1) and the Digital transformation handbook for primary health care: Optimizing person-centred point of service systems (7) provide additional guidance.

Step 5: monitoring, evaluation and scale-up

To support wider scale-up and rollout at the country level, rigorous monitoring and evaluation of the digital systems that have leveraged SMART guidelines should be conducted to assess acceptability and feasibility, as recommended for any digital health intervention. The WHO Monitoring and evaluating digital health interventions: A practical guide to conducting research and assessment provides additional guidance on this (38).

Part 2.

Digital adaptation kit content for birth defects surveillance



Component

1

Health interventions and recommendations



This DAK focuses on the following health interventions and recommendations related to birth defects surveillance.

WHO universal health coverage interventions referenced in this DAK

The key interventions for birth defects surveillance referenced in this DAK, as defined in the WHO UHC compendium (24) and WHO birth defects surveillance guidelines and guidance documents, are:

- screening and diagnosis of congenital anomalies by means of physical examinations;
- laboratory tests; and
- imaging studies.

Digital health interventions covered in this DAK

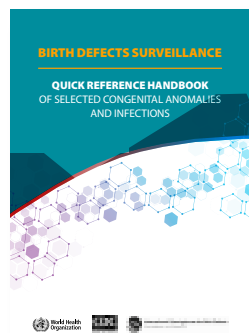
Table 2 lists the digital health interventions incorporated into this DAK, extracted from the WHO Classification of digital interventions, services and applications in health (3).

Table 2. Digital health interventions incorporated into the DAK for birth defects surveillance

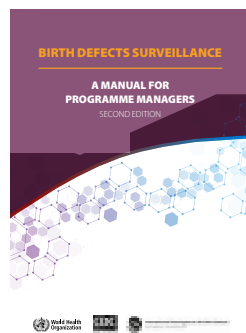
Category	Digital health intervention
1.8 Person-centred consent management	1.8.1 Manage provision and withdrawal of consent by individual(s)
2.1 Identification and registration of persons	2.1.1 Verify a person's unique identity
2.2 Person-centred health records	2.2.2 Manage person-centred structured clinical records 2.2.3 Manage person-centred unstructured clinical records (e.g. notes, images, documents) 2.2.4 Routine health indicator data collection and management
2.3 Healthcare provider decision support	2.3.2 Provide checklist according to protocol
2.5 Healthcare provider communication	2.5.3 Transmit routine news and workflow notifications to healthcare provider(s)
2.8 Healthcare provider training	2.8.1 Provide training content to healthcare provider(s)
2.10 Laboratory and diagnostics imaging management	2.10.3 Capture diagnostic results from digital devices
4.1 Data management	4.1.2 Data storage and aggregation 4.1.3 Data synthesis and visualization
4.2 Data coding	4.2.1 Parse unstructured data into structured data 4.2.3 Classify disease codes or cause of mortality
4.5 Data governance compliance	4.5.1 Authentication and authorization 4.5.2 Data privacy protection 4.5.3 Data consent and provenance

WHO guidelines, recommendations and guidance

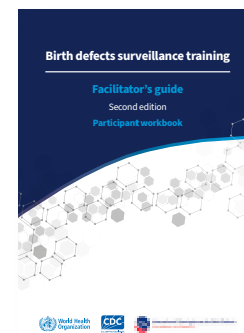
The DAKs are intended to reflect health recommendations and content that has already been published in WHO guidelines and guidance documents. The health content and interventions for this DAK are based on the following WHO documents:



Birth defects surveillance: Quick reference handbook of selected congenital anomalies and infections (28).



Birth defects surveillance: A manual for programme managers, 2nd edition (23).



Birth defects surveillance training: Facilitator's guide, 2nd edition (29).

Other WHO guidance represented in the DAK includes:

- Congenital disorders (39);
- WHO recommendations: Optimizing health worker roles to improve access to key maternal and newborn health interventions through task shifting (40);
- UHC compendium: Repository of interventions for universal health coverage (24);
- WHO UHC service planning, delivery & implementation (SPDI) platform (25); and
- Family planning: A global handbook for providers – Evidence-based guidance developed through worldwide collaboration, 3rd edition (41).

Guidance represented in the DAK from non-WHO sources includes:

- Annual report 2014 from the International Clearinghouse for Birth Defects Surveillance and Research (ICBDSR) (42);
- Birth defects information from the US Centers for Disease and Prevention (CDC) (43); and
- Guidelines for data registration from the European Surveillance of Congenital Anomalies (EUROCAT) (44).

Component

2

Generic personas



A persona is a depiction of a relevant stakeholder or end-user of the system. Specific roles and demographic profiles of personas vary, depending on the setting, but generic personas are based on the WHO core competencies and credentials of different health worker personas. Note that these are developed based on synthesis across multiple contexts as a starting point, and further contextualization will be required according to the needs, motivations and challenges of the key personas in each setting.

Targeted generic personas

The targeted personas for the birth defects surveillance DAK are health workers operating in care settings that are able to provide the required essential interventions for birth defects surveillance. In this case, the targeted personas are data collectors, birth defects specialists, surveillance programme managers and surveillance data managers for the digital client health record and decision-support systems. In the health systems surveyed for this DAK, the common combination of service providers was data collectors, birth defects specialists and surveillance professionals. The key competencies of data collectors, birth defects specialists, surveillance programme managers and surveillance data managers are defined by WHO and listed in Table 3 (45).

Table 3. Targeted generic personas for the birth defects surveillance DAK

Name	Description	Different names	International Standard for Classification of Occupations (ISCO) code ^a
Data collector	A wide variety of health workers may be involved in collecting or abstracting data on birth defects.	Health extension worker, community health worker.	3253 Community health workers
		• Nurse • Midwife • Auxiliary nurse midwife.	2221 Nursing professionals, 2222 Midwifery professionals, 3221 Nursing associate professionals, 3222 Midwifery associate professionals.
		Family doctor, general practitioner, medical doctor, nonspecialist doctor, specialist doctor (e.g. gynaecologist, obstetrician, paediatrician).	2211 Generalist medical practitioners, 2212 Specialist medical practitioners.
		Clerk or receptionist at hospital admissions.	4229 Client information workers not elsewhere classified (clerk, hospital admissions)
		Data clerk, data abstractor.	4132 Data entry clerks
Birth defects specialist	Physicians, geneticists or other specialists that have knowledge or training in the identification of birth defects. In some birth defects surveillance systems, multiple birth defects specialists review the birth defect diagnosis, either individually or together in a panel; in this DAK, the singular form of “birth defects specialist” is used throughout, but some implementations may involve multiple birth defects specialists.	• Birth defects specialist, physicians (specialists); and • Geneticist.	2212 Specialist medical practitioners (physician, specialist) Not available
		Nurse	2221 Nursing professionals
Surveillance programme manager	Surveillance programme managers may be responsible for developing and implementing public health interventions to reduce the incidence of birth defects. Surveillance programme managers may use data from surveillance programmes to inform decisions about public health interventions and policies related to birth defects and health-care needs. Surveillance programme managers are responsible for training data collectors and data managers to ensure the accuracy and timeliness of birth defect identification and surveillance procedures.	Physician, specialist.	2212 Specialist medical practitioners (physician, specialist)
		Family doctor, general practitioner, medical doctor, specialist doctor (e.g. gynaecologist, obstetrician), nonspecialist doctor.	2211 Generalist medical practitioners
		Epidemiologist or other public health officer.	Not available
		District health manager, health management information systems focal point, monitoring and evaluation focal point, facility supervision manager, district health information officer, national monitoring and evaluation officer.	1342 Health service managers

Name	Description	Different names	International Standard for Classification of Occupations (ISCO) code ^a
Surveillance data manager	Medical records and health information technicians develop, maintain and implement health records processing, storage and retrieval systems in medical facilities and other health-care settings to meet the legal professional, ethical and administrative records-keeping requirements of health services delivery. Surveillance data managers may be responsible for overseeing the collection and analysis of data on birth defects and using this information to identify trends and potential risk factors. They may perform some of the functions of the surveillance programme data manager in terms of preparing reports. They must have information technology (IT) and data management skills because they are also responsible for IT and data system administrative tasks related to birth defects surveillance, such as creating and maintaining users.	Health associate professional	3252 Medical records and health information technicians
		Data manager	2521 Database designers and administrators

^a International Standard Classification of Occupations (ISCO). Geneva: International Labour Organization (46).

Related personas

In addition to the targeted personas described above, there may be value in exploring other health worker occupational groups and personas within the context of birth defects surveillance (Table 4).

Table 4. Related personas for the birth defects surveillance DAK

Name	Description	Different names	International Standard for Classification of Occupations (ISCO) code ^a
Newborn	An infant aged under 28 days. Up to the age of 5 years, a child is at the highest risk of dying during the first 28 days of life. Most newborn deaths take place in developing countries, where access to health care is low.	• Neonate • Baby • Infant • Child.	Not applicable
Mother	A woman who has given birth to a newborn baby with a possible birth defect.		Not applicable

Name	Description	Different names	International Standard for Classification of Occupations (ISCO) code ^a
Informant	A person providing information about the mother or newborn. In the majority of cases, the informant is the mother, but there may be situations where the mother is unavailable or has died; in these cases, family members or friends may be responsible for providing information.	• Mother • Father • Parent • Grandmother • Grandfather • Family member • Friend • Neighbour.	Not applicable

^a International Standard Classification of Occupations (ISCO). Geneva: International Labour Organization (46).

Additional considerations for contextualizing personas

This section provides an overview of the generic roles of the targeted personas. It is important to contextualize these personas to the setting. The generic personas described in Tables 3 and 4 can be supplemented by reflecting on:

- **Background and demographics**, such as gender, age, whether they are from the community, familiarity with digital devices, and whether they have access to a mobile phone or smartphone;
- **Local environment** and any relevant contextual information about their surroundings, such as work-site characteristics, rural/urban location, availability of electricity, water and internet, and distance from the nearest referral facility;
- **Expected roles and responsibilities** based on the country context, and how these differ from the roles and responsibilities defined by WHO;
- **Actual roles and responsibilities**, and whether there is any difference from the expected roles and responsibilities;
- **Context**, such as internet connectivity, how personas are compensated, distance from the nearest referral facility, and which other personas or health workers they interact with;
- **Challenges**: The day-to-day challenges the end-user may face;
- **Motivations**, such as what success looks like to them, and whether there are targets they need to achieve.

Component

3 User scenarios



User scenarios are a narrative description of how the end-user could interact with the digital system. The user scenario permits a better understanding of how the system will be used and how it could fit into existing workflows. The user scenario provides context in a storytelling format. Within the user scenario, it should be possible to derive the key components that are further elaborated in the rest of the DAK. This includes the core data elements and core functionality of a digital system that would be needed. Although there is no clear template for a user scenario, it should:

- include personas involved based on the generic personas component;
- have a narrative description of who and how a digital system would be used;
- provide details on what kinds of data are collected and what kinds of decisions are made; and
- reflect the workflows that are further elaborated in [Component 4: generic business processes and workflows](#).

How to interpret user scenarios

User scenarios are helpful tools for better understanding the context in which a digital tool would operate. They offer insights into the key data elements that need to be recorded and accounted for in the database. The context in which the tool would operate, illuminated by user scenarios, provides insight into the functional and non-functional requirements that the system would need. In the example user scenario below, key data elements that should be recorded or calculated are highlighted **in yellow**. Key functional and non-functional requirements that should be included in the system are highlighted **in green**.

Table 5 shows the interpretation of the example user scenario “Community health worker follows up on report of mother of newborn with birth defect”.

Table 5. Interpretation of the user scenario “Community health worker follows up on report of mother of newborn with birth defect”

Data elements to be collected	Functional and non-functional requirements
- Mother’s first name; - Mother’s family name or last name; - Mother’s primary residential address at time of delivery; - First name; - Last name; - Date of birth; - Case record ID; - Signature for consent; - Sex; - Gestational age at birth; - Weight; - Length; - Mother’s age; - Total number of previous live births; - Father’s age; - Parents related; - Congenital anomaly clinical presentation; - Congenital anomaly diagnosis; - Congenital anomaly diagnosis status; - Congenital anomaly description; - Photograph.	- Ability to create new case records; - Ability to assign unique IDs; - Ability to retrieve case record; - Ability to receive digital signature; - Ability to gather mother’s contact and address information; - Ability to provide interactive series of illustrations, photographs and descriptions to help with diagnosis; - Ability to take, store and attach digital photographs to case record; - Ability to submit case record to birth defects surveillance system remotely.

User scenario: community health worker follows up on report of mother of newborn with birth defect

Key personas

Mother: Hope

Newborn: Mosi

Data collector: Aisha

Aisha, a community health worker, has been notified by the local birthing facility of a mother (Hope) requesting a home visit from a community health worker. Hope and baby Mosi were discharged home two days ago. Aisha notes that the reported concern from Hope is that milk is coming out of Mosi's nose during breastfeeding. The notification includes the **mother's first and last name**, the **newborn's name**, the **address**, and the **newborn's date of birth**. Aisha opens her birth defects surveillance mobile data collection application on her phone and **creates a new case record**. She enters Hope and Mosi's names, Mosi's date of birth and the address. The **application assigns a unique case record ID number**.

Aisha travels to Hope's home to meet Hope and Mosi. She explains the purpose of her visit. With Hope's permission, Aisha begins her examination of Mosi. Based on the reported concern, Aisha does a careful examination of Mosi's head and identifies an opening in the roof of Mosi's mouth. Recognizing that this is a birth defect, **she retrieves the case record in the mobile application**. In the health district, the ministry of health requires informed consent to be included in the birth defects surveillance system. Before she can continue with data collection, Aisha reads the consent text from the mobile application to Hope. The surveillance protocol allows for digital signatures for **consent**, so **Hope signs directly in the application**. Aisha begins collecting general information about Mosi and the birth (e.g. **sex, gestational age at birth, weight, length**) and information regarding Hope and Mosi's father (e.g. **mother's age, total number of previous live births, father's age, parents related**).

The **application directs Aisha through the questions detailing the birth defect** through an **interactive series of illustrations, photographs and descriptions** of birth defects based on the WHO Quick reference handbook of selected congenital anomalies and infections (28). Following the application's directions and her own assessment, Aisha selects from a dropdown list the **congenital anomaly diagnosis** of "cleft palate" and the **congenital anomaly clinical presentation** as being "isolated". She assigns a **congenital anomaly diagnosis status** of "possible" and enters a **text description of the congenital anomaly** in the birth defects surveillance mobile data collection application. Aisha **takes photographs** of Mosi's head and open mouth **with her mobile phone via the application**, making sure the opening is clearly visible, as recommended by the application based on the Quick Reference Handbook.

Having completed data collection, Aisha conducts her other responsibilities as a community health-care worker on a newborn visit and follows local guidelines for clinical referral. Aisha thanks Hope and **submits the case record in the birth defects surveillance mobile data collection application**.

Corresponding business processes
(see [Component 4: generic business processes and workflows](#))

This scenario refers to the following business processes:

A. Data collection and abstraction

User scenario: birth defect data abstraction by data collector in ward

Key personas	Newborn: <i>Adaeze</i> Data collector: <i>Paul</i>
	Paul, a nurse in a birthing facility, has been requested to assist in a newborn examination of a baby (Adaeze) with a possible birth defect, given his experience with submitting cases to the birth defects surveillance system. Paul joins the examination and sees an unusual area on the baby’s back. Paul suspects this to be a case of spina bifida. The protocol for birth defects surveillance in the region does not require consent from a parent to include information about the birth in the surveillance system. Therefore, Paul has autonomy to create a new record for Adaeze directly from the facility’s health records. Paul opens the birth defects surveillance mobile data collection application on his phone and creates a new case record with a unique case record ID. The birthing facility uses an HMIS for its client records and Paul is able to electronically transfer the existing information regarding Adaeze, his birth and his parents into the birth defects surveillance system application on his phone. In the section concerning details of the birth defect, Paul follows the application’s guidance and documentation checklist, which is based on the WHO Quick reference handbook of selected congenital anomalies and infections (28). Paul knows that spina bifida occurs along the spine, but he is unsure if the lesion on Adaeze’s back is in the correct location to qualify. Not being completely sure of his diagnosis, Paul assigns the congenital anomaly diagnosis as “spina bifida” and the congenital anomaly diagnosis status as “possible”. He takes photographs of Adaeze’s back with his mobile phone via the application to help the birth defects specialist make an accurate diagnosis. Having completed data collection, Paul submits the case record to the birth defects surveillance system.

Corresponding business processes (see Component 4: generic business processes and workflows)	This scenario refers to the following business processes: A. Data collection and abstraction
--	--

User scenario: birth defects specialist reviewing and confirming case

Key personas

Surveillance data manager: *George*

Data collector: *Atieno*

Birth defects specialist: *Onyango*

Newborn: *Adaeze*

Onyango, the birth defects specialist for the region's birth defects surveillance system, receives occasional birth defect case reports to review for inclusion in the surveillance system. He **logs into the data portal for the birth defects surveillance system on his laptop** and finds there is only one case to review **in his queue** – a report from data collector nurse Atieno about Adaeze, a baby with spina bifida.

Onyango **reviews the submitted photographs** and is able to diagnose the spina bifida as thoracic spina bifida – a birth defect he knows to be included in the surveillance system. Based on his clinical training, Onyango is aware of the spina bifida sequence and is able to rule out hydrocephalus, because the submitted photographs captured the baby's head. Onyango is unable to rule out talipes equinovarus, because the baby's feet are not captured in the photographs.

Onyango **updates the case record** to request further information from the data manager, specifically requesting for **photographs to be taken of Adaeze's feet**. Onyango **returns the case report to George, the data manager**.

George **receives a notification for more information in his work queue**, **reviews** it and **assigns it to nurse Atieno** for follow-up. Atieno receives the notification and travels to the ward to find Adaeze. Atieno takes the requested photographs of the baby's feet and submits them to the birth defects surveillance system.

Once he has received and reviewed the case, George returns the case to Onyango, who reviews the new photographs of Adaeze's feet and determines that Adaeze has normal feet, which rules out talipes equinovarus. Therefore, Onyango does not need to edit his previously entered diagnosis. Onyango completes his review and submits the record to George. George **notifies the care provider of the updated diagnosis**.

Corresponding business processes
(see [Component 4: generic business processes and workflows](#))

This scenario refers to the following business processes:

B. Case confirmation

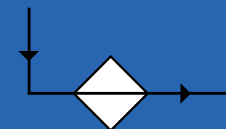
User scenario: birth defects surveillance data manager preparing monthly reports

Key personas	Surveillance data manager: <i>Nasir</i> Surveillance programme manager: <i>Fatima</i>
	Nasir is the surveillance data manager for his health district birth defects surveillance system. At the end of every January and July, he is required to prepare a report of the birth defects that occurred in the district in the preceding six months. As it is the end of July, Nasir retrieves data specific to his district from the birth defects surveillance system, including the birth denominators that have been collected from the beginning of the year through to the end of June. He checks the data errors on the screen, including looking for cases that do not meet the birth defect case definition, date transpositions and data outliers. He finds a couple of minor date errors and makes corrections to the data. Nasir cleans, deidentifies and aggregates the data, (prepares the database for analysis) and generates the half-year report. He sends the report to Fatima, the district birth defects surveillance programme manager. Fatima reviews the report for issues. She notices that one of the health facilities has a case count that is too high for the size of the facility. She makes comments and sends the report back to Nasir. Nasir reviews the issue and sees that two similarly named facilities were switched. Nasir makes corrections to the data, regenerates the report, and sends it to Fatima. Fatima reviews the new report. Finding the corrections acceptable, she writes her summary text, providing a high-level synthesis of the prevalence statistics and any notable trends. She adds her summary text to the report and approves the report with her digital signature and submits it to the birth defects surveillance system. The birth defects surveillance system automatically emails the report to the distribution list of health facilities in the surveillance system in the district and ministry of health officials and makes it available on the district data portal.
Corresponding business processes (see Component 4: generic business processes and workflows)	This scenario refers to the following business processes: C. Data aggregation and dissemination

Component

4

Generic business processes and workflows



A business process is a set of related activities or tasks performed together to achieve the objectives of the health programme area, such as registration, counselling and referrals (1,2). Workflows are a visual representation of the progression of activities (tasks, events, interactions) that are performed within the business process (2). The workflow provides a story for the business process being diagrammed and is used to enhance communication and collaboration among users, stakeholders and engineers.

This DAK focuses on key business processes conducted by the personas (described in Component 2: generic personas) as part of birth defects surveillance. These business processes are listed in Table 6. For each business process, the corresponding business processes and data elements are detailed in the following sections of this document.

Table 6. Overview of key business processes^a

Corresponding letter for process	Process name	Process ID ID used to reference this process throughout the DAK	Personas Individuals interacting to complete the process	Objectives Concrete statement describing what the process seeks to achieve	Task set General set of activities performed within the process
A	Data collection and abstraction	BDS.A	• Mother • Newborn • Informant • Data collector.	To register all birth defect case data in the system.	Starting point: the data collector receives a notification of newborn with a possible birth defect. In a health facility-based surveillance system, this could happen automatically through the facility HMIS, or the data collector could be present on the delivery ward or be notified through other means. In a community health-based surveillance system, the data collector could be directed to the mother and newborn as part of community outreach activities or regular follow-up visits to pregnant women or other forms of notifications • Search for client record; • Review and update client record; • Create a new client record; • Collect all available data related to the newborn, mother and father, and details related to the birth defect(s), including photographs if possible.

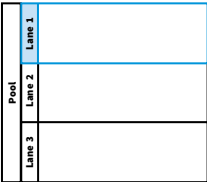
Corresponding letter for process	Process name	Process ID ID used to reference this process throughout the DAK	Personas Individuals interacting to complete the process	Objectives Concrete statement describing what the process seeks to achieve	Task set General set of activities performed within the process
B	Case confirmation	BDS.B	• Data collector • Surveillance data manager • Birth defects specialist.	To confirm or correct the birth defect diagnosis.	Starting point: birth defect case has been completed by the data collector and submitted to the birth defects surveillance system • The surveillance data manager reviews individual cases; the case is sent back to the data collector if there are any issues; • The surveillance data manager sends cases to the birth defects specialist; • The birth defects specialist reviews each case; if the correct information is not available to review the case record, the case is sent back to the data manager and then to the data collector for more information; • The birth defects specialist confirms, updates or discards diagnosis and updates case records; • Case records are sent back to the data manager for final review.
C	Data aggregation and reporting	BDS.C	• Surveillance data manager • Surveillance programme manager.	To aggregate client-level data into validated aggregate reports, use the data and submit reports at the facility level.	Starting point: on a scheduled or ad hoc basis, the surveillance data manager reviews the data from a specific time period and surveillance population • Check data quality; • Correct fixable errors; • Generate and review aggregate reports or databases; • Submit for approval; • Provide feedback and any changes required.

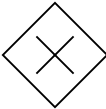
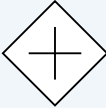
^a The business processes described in the DAK are those listed in this table. Processes that are part of birth defects surveillance but not included in this DAK include billing and clinical care and treatment (shaded grey in Fig. 4). These processes may be required but are highly country- and context-specific. If applicable, the billing business process could include an insurance or coverage check, which could take place at any one of many points during a birth defects surveillance visit. All the processes that are part of birth defects surveillance are shown in Fig. 4. The editable files of each business process in .bpmn format can be found here: <https://smart.who.int/dak-bds/business-processes.html>.

Overview of key processes

This section illustrates the workflows of the identified processes using standardized notations for business process mapping, based on the business process model and notation (BPMN) standard (47). Table 7 provides an overview of this notation.

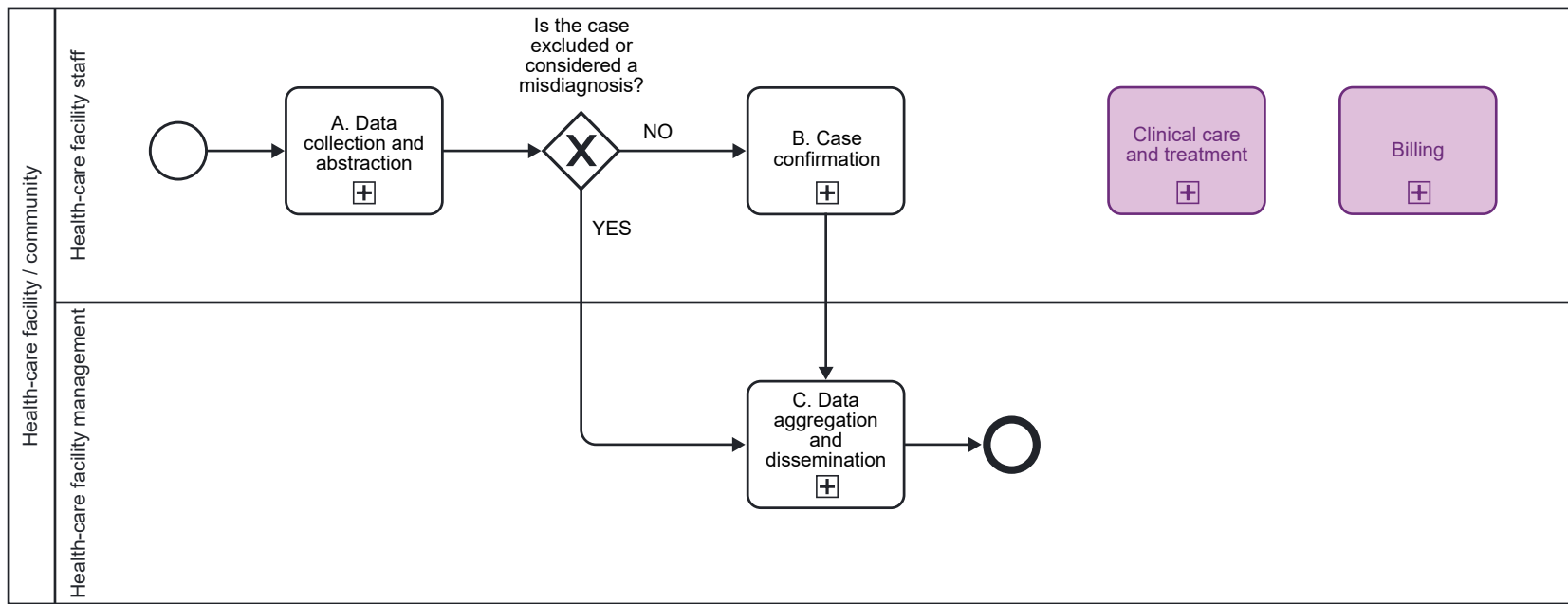
Table 7. Business process symbols and notation used in workflows

Symbol	Name	Description
	Pool	The pool assumes process control and assigns tasks. The pool consists of one or more “swim lanes” that depict all the individuals or types of user involved in carrying out the business process or workflow. Diagrams should be clear, neat and easy for all viewers to understand the relationships across the different swim lanes. For example, a pool could depict the business process of conducting an outreach activity, which involves multiple stakeholders represented by different swim lanes. A business process diagram can contain two or more pools that connect through message flows (messages received from outside the pool) – this indicates different process owners collaborating and exchanging information. For example, in the case of a referral, Health Facility A (the referring facility) represents the process owner that internally assigns the tasks for the actors represented as swim lanes. Health Facility B (the receiving facility) has allocated its own pool and is considered a different process owner with the same role. The two facilities exchange information relevant for a referral (e.g. client history, referral details), represented as message flows, but each facility has control only over the internal activities, without directly impacting the activities of the other facility.
	Swim lane	Each individual or type of user is assigned to a swim lane, a designated area for noting the activities performed or expected by that specific actor. For example, the birth defects surveillance health worker may have one swim lane, the supervisor another swim lane, and the clients another swim lane.
	Start event or trigger event	The workflow diagram should contain a start event and an end event, defining the beginning and completion of the task.
	End event	Multiple end events can be depicted across multiple swim lanes in a business process diagram. For clarity, there should be only one end event per swim lane.
	Activity, process, step or task	Each activity should start with a verb, such as “Register client” or “Calculate risk”. Between the start and end of a task, there should be a series of activities noting the successive actions performed by the actor for that swim lane. There can be subprocesses for each activity.
	Activity with subprocess	This denotes an activity that has a much longer subprocess, which is detailed in another diagram. If the diagram is too complex, the subprocess symbol should be used to reference another process depicted on another page.

Symbol	Name	Description
	Sequence flow	This denotes the flow direction from a process to the next process. The end event should not have any output arrows. All symbols (except for the start event) may have an unlimited number of input arrows. All symbols (except for the end event and gateway) should have one and only one output arrow, leading to a new symbol, looping back to a previously used symbol or to the end event symbol. Connecting arrows should not cross over each other.
	Exclusive gateway	This depicts a fork or decision point in the workflow. This may be a simple binary (e.g. yes/no) filter with two corresponding output arrows, or a different set of outputs. Only two outputs should originate from a decision point. If more than two output or sequence flow arrows are needed, you are likely trying to depict decision-support logic or a business rule, which should be depicted as an activity with a business rule instead (see above).
	Parallel gateway	This is used to model concurrency in a process. This type of gateway allows forking into multiple paths of execution or joining multiple incoming paths of execution. An important difference between this and other gateway types is that the parallel gateway does not evaluate conditions.
	Throw-link event	This serves as the start of an off-page connector. It is the end of the process when there is no more room on the page for that workflow. It is the end of a process on the current page or the end of a subprocess that is part of a larger process. A catch-link event must follow a throw-link event.
	Catch-link event	This serves as the end of an off-page connector. It is the start of a new process on a different page from the throw-link event or the start of a subprocess that is part of a larger process. A throw-link event must be aligned to a catch-link event.
	Loop activity	This indicates that the activity repeats until a defined condition applies or ceases to apply. The condition on which a loop executes is included as an annotation.

^a For description of key processes, see Table 6.

Fig. 4 Overview of key birth defects surveillance processes^a



^a For description of key processes, see Table 6.
Note: the purple boxes are not covered in this DAK.

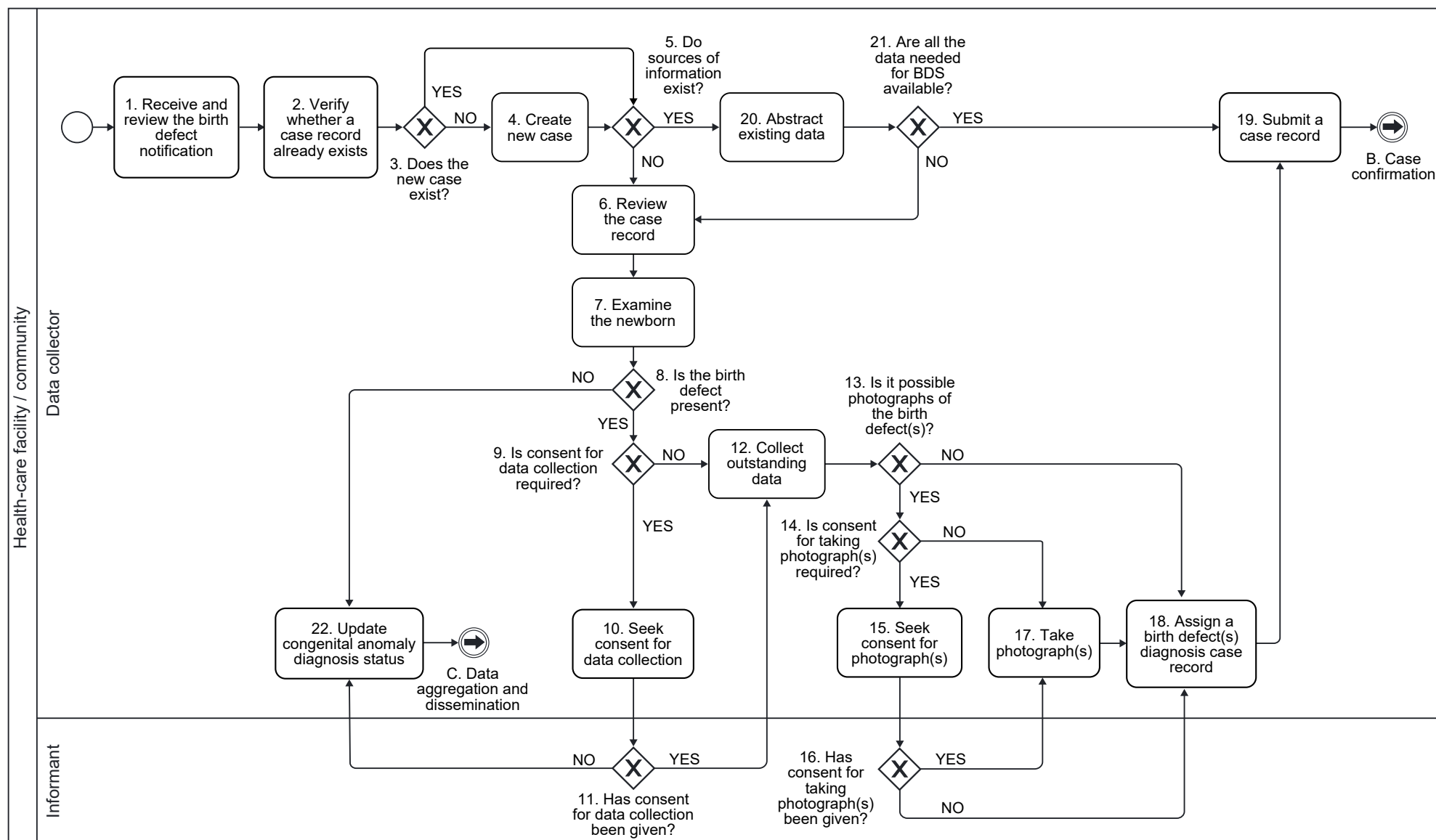
Workflows

Workflows represent the progression of activities performed within the business process. They help users and stakeholders understand the relationship between activities, data elements and decision-support needs. The workflows shown here depict processes that have been generalized and may not reflect variation and nuances across different settings. The simplicity of the workflow may not adequately illustrate nonlinear steps that may occur.

Workflow BDS.A: business process for data collection and abstraction

Objective: to register all birth defect case data into the system (Fig. 5).

Fig. 5 Workflow BDS.A: business process for data collection and abstraction



DATA COLLECTION AND ABSTRACTION BUSINESS PROCESS NOTES AND ANNOTATIONS

General note: data collection and abstraction is the process in which the data collector collects data concerning the case directly from the informant or abstracts data from existing medical records. These activities can be performed in the health-care facility or in the community (e.g. client's home, workplace, mobile van), depending on where the encounter takes place:

1. Receive and review the birth defect notification:

- Receive and review notification of newborn with possible birth defect(s).
- The notification may be a verbal notification, paper form or email message, or a sophisticated notification from an advanced HMIS that already incorporates all the information needed.

2. Verify whether a case record already exists:

- The data collector searches the system to verify whether a case record for the newborn has already been created in the system. If so, the data collector continues the workflow using the existing case record.
- If the search result contains multiple matches, the data collector selects the best match and indicates the other returned results as possible duplicates, so the surveillance data manager can deduplicate the cases in Business process C: data aggregation and dissemination.

3. Does the case exist?

- Does a case record for the newborn already exist?

4. Create new case:

- The data collector creates a new case in the birth defects surveillance system, and a case record ID is automatically assigned.

5. Do sources of information exist?

- Do medical records or other sources of information exist for the newborn or mother?

6. Review the case record:

- If there are no sources of information for the newborn or mother, or information exists but is not sufficient to submit the case record, the data collector reviews and assesses the extent of the missing required data for case submission before starting the physical examination of the newborn.

7. Examine the newborn:

- The data collector preforms a visual inspection of the newborn to confirm the presence of eligible birth defects or to identify a misdiagnosis and mistaken notification.

8. Is a birth defect present?

- Is a birth defect determined to be present after visual inspection of the newborn?

9. Is consent for data collection required?

- When birth defects are confirmed, the data collector may need to ask for consent, according to local or national policies, before proceeding to data collection.

10. Seek consent for data collection:

- Whether the consent is documented in writing or digitally or only witnessed depends on local practice (signed consent or witnessed consent if the child's parent or legal guardian is illiterate).

11. Has consent for data collection been given?

- Has consent for data collection been given based on local practice and the locally applicable legal framework?

12. Collect outstanding data:

- Information relevant to the case is collected from the informant (e.g. date of birth, sex, weight, length).

13. Is it possible to take photograph(s) of the birth defect(s)?

- The data collector assesses whether it is possible to take photograph(s) of the birth defect(s) for case confirmation.

14. Is consent for taking photograph(s) required?

- If taking birth defect(s) photograph(s) is possible, the data collector may need to ask for consent, according to national policies.

15. Seek consent for photograph(s):

- Whether the consent is documented in writing or digitally or only witnessed depends on local practice (signed consent or witnessed consent if the child's parent or legal guardian is illiterate).

16. Has consent for taking photograph(s) been given?

- Has consent for taking photograph(s) been given based on local practice and the locally applicable legal framework?

17. Take photograph(s):

- The data collector takes photograph(s) and includes them in the case record. See Appendix J in Birth defects surveillance: A manual for programme managers, 2nd edition (23).

18. Assign a birth defect(s) diagnosis:

- The data collector makes their determination regarding a possible birth defect(s) diagnosis.
- If possible, birth defects surveillance should provide interactive tools to assist with the assignment of birth defects diagnoses and ICD codes using photographs and detailed birth defects information.
- If the data collector is also a birth defects specialist or a medical professional with appropriate training, then assigning a confirmed diagnosis is acceptable. See Birth defects surveillance: Quick reference handbook of selected congenital anomalies and infections (28).

19. Submit a case record:

- The data collector submits the case record, and the workflow continues with Business process B: case confirmation.

20. Abstract existing data:

- If medical records or other sources of information exist for the newborn or mother, the data collector abstracts or imports all existing information.

21. Are all the data required for birth defects surveillance available?

- If the case record contains all the necessary information for submission, the workflow continues with Step 19: submit a case record.
- If the existing information is not enough to submit the case record, the workflows continues with Step 6: review the case record.

22. Update congenital anomaly diagnosis status:

- If the examination of the newborn does not confirm the presence of eligible birth defects, the data collector sets the congenital anomaly diagnosis status to “misdiagnosis” (if diagnosis has already been made).
- The status is set to “excluded” if the birth defect(s) present does not meet the inclusion criteria for the list of birth defects under surveillance, as set by the local or national government.
- The data collector marks the case record as complete. The process continues with Business process C: data aggregation and dissemination.

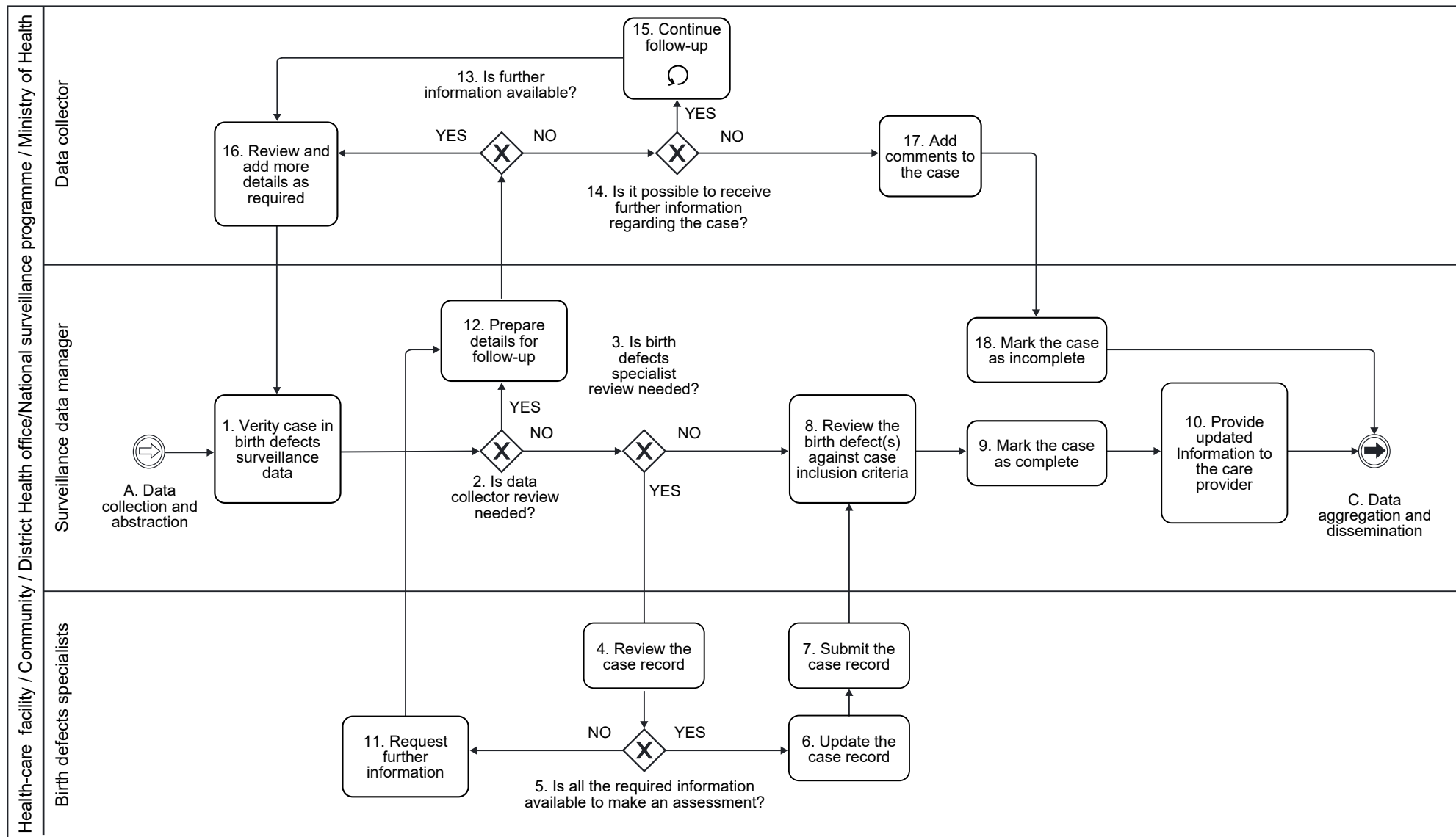
For guidelines and guidance, see:

- Birth defects surveillance: A manual for programme managers, 2nd edition (23).
- Birth defects surveillance: Quick reference handbook of selected congenital anomalies and infections (28).
- Birth defects surveillance training: Facilitator’s guide, 2nd edition (29).

Workflow BDS.B: business process for case confirmation

Objective: to confirm or correct the birth defect diagnosis (Fig. 6).

Fig. 6 Workflow BDS.B: business process for case confirmation



CASE CONFIRMATION BUSINESS PROCESS NOTES AND ANNOTATIONS

General notes: case confirmation is the process by which birth defects specialists and data managers review the clinical characteristics of cases of birth defects to approve their inclusion in the surveillance system and the accuracy of diagnoses. In addition to a case having at least one of the selected birth defects that the surveillance system is capturing, other clinical characteristics or inclusion criteria may include age, pregnancy outcome, gestational age at delivery, birth weight or maternal residence. More information on defining these inclusion criteria and examples can be found in Birth defects surveillance: A manual for programme managers, 2nd edition (23). These activities can be taken at the community, health facility, district/area/regional health office level or at the national level with the ministry of health or similar national organization.

1. Verify case in birth defects surveillance data:

- The surveillance data manager verifies the case in the birth defects surveillance data and ensures there are no duplicates. In some cases, the same child is diagnosed with birth defects by different teams and at different points in time, and different case records with unique IDs are generated. Care should be taken to not confuse record duplication with births involving twins or other multiple births with birth defects. If multiple records are found for the same newborn, consider merging or deleting duplicate records, according to the HMIS guidelines:
 - Correct birth defect(s) ICD code(s) is/are assigned; and
 - The data are complete.

2. Is data collector review needed?

- The surveillance data manager reviews the case for potential data entry errors and returns the case to the data collector for review if needed.

3. Is birth defects specialist review needed?

- The surveillance data manager follows the local or national protocol to establish whether review by a birth defects specialist is needed. This may include checking that all birth defects in the case record have the congenital anomaly diagnosis status set to “confirmed”, and that birth defects that require review by a birth defects specialist, according to protocol, are identified.

- A fully trained health worker may set the diagnosis status to “confirmed” while performing the data collection and abstraction for the case record. In this situation, the process continues with Step 8: review the birth defect(s) against case inclusion criteria. If review by a birth defects specialist is needed, the process continues with Step 4: review the case record.

4. Review the case record:

- The case record, including available photograph(s) of the birth defect(s), is reviewed.
- The birth defects specialist uses their clinical judgement and the Quick reference handbook of selected congenital anomalies and infections (28) to assess the case record diagnosis and ICD codes.

5. Is all the required information available to make an assessment?

6. Update the case record:

- Based on clinical judgement, update case record information, such as:
 - congenital anomaly description;
 - congenital anomaly diagnosis status;
 - congenital anomaly clinical presentation;
 - congenital anomaly diagnosis; and
 - ICD codes.
- When familiar with case inclusion criteria, the birth defects specialist updates the congenital anomaly diagnosis status based on the inclusion criteria

determination. The congenital anomaly diagnosis status is set to “excluded” for cases that do not meet inclusion criteria. Familiarity with the case inclusion criteria is determined based on the national protocol or location-specific guidance.

7. Submit the case record:

- Submit the case record for review by the surveillance data manager.

8. Review the birth defect(s) against case inclusion criteria:

- The birth defects are reviewed against case inclusion criteria. Birth defects that do not meet the inclusion criteria, according to the protocol, are excluded and the case record is updated.

9. Mark the case as complete:

- The birth defect case report is marked as completed once the final case diagnoses have been completed and the case inclusion criteria for the case have been assessed.

10. Provide updated information to the care provider:

- The surveillance data manager provides the updated information to the care provider with the purpose of communicating this information further to the newborn’s mother or legal guardian. This may be the same data collector that initially created the case record or another health-care provider that interacts with the newborn’s mother or legal guardian.
- For guidelines and guidance, see Appendix F in Birth defects surveillance: A manual for programme managers, 2nd edition (23).

11. Request further information:

- The birth defects specialist specifies the missing information that is required to make an assessment and returns the case record to the surveillance data manager. The birth defects surveillance system could trigger a notification

with the purpose of informing the surveillance data manager about the need for further information on the case.

12. Prepare details for follow-up:

- The surveillance data manager adds more information to the case record, if needed, specifying what information is missing and requesting the information from the data collector.

13. Is further information available?

- The health-care provider that takes over the case to assess whether further information can be provided is not necessarily the same person who created the case record.

14. Is it possible to receive further information regarding the case?

15. Continue follow-up:

- If it is possible to get more information regarding the case, the data collector communicates with the mother or legal guardian, physicians or medical records and retrieves that information. The national protocol or location-specific guidance should specify how long for or how many times to follow up.

16. Review and add more details as required:

- If required information is available, the data collector updates the case record with it. The workflow continues with Step 1: verify case in birth defects surveillance data.

17. Add comments to the case:

- If further information is not available and it is not possible to receive it from the mother or the legal guardian, the data collector adds comments to the case describing the inability to retrieve and provide the needed information.

18. Mark the case as incomplete:

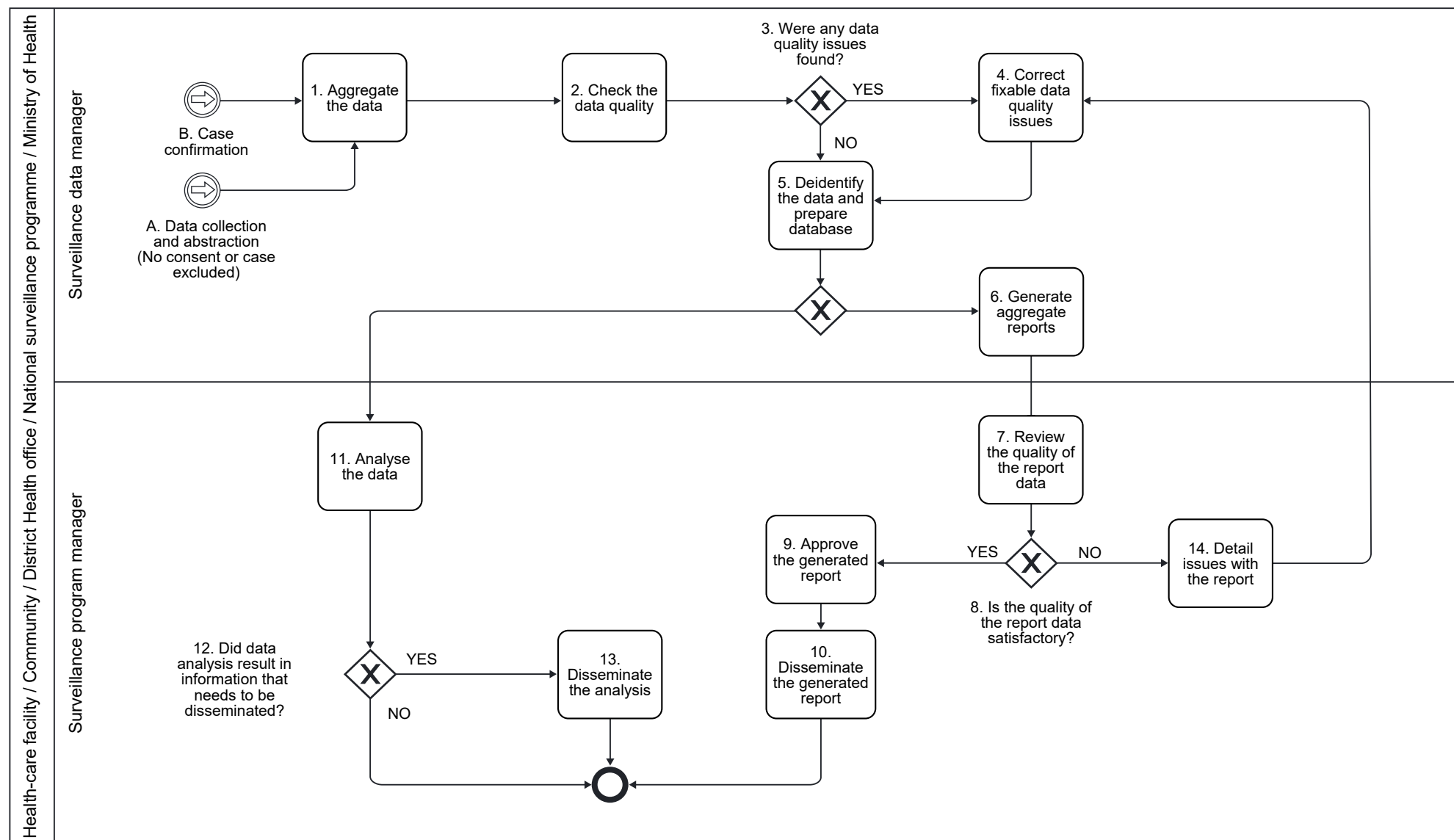
- The surveillance data manager marks the case as incomplete due to the lack of required information. The workflow continues with Business process C: data aggregation and dissemination.

For guidelines and guidance, see Birth defects surveillance: Quick reference handbook of selected congenital anomalies and infections (28).

Workflow BDS.C: business process for data aggregation and dissemination

Objective: to aggregate client-level data into validated, aggregate reports, analyse the data and disseminate the results (Fig. 7).

Fig. 7 Workflow BDS.C: business process for data aggregation and dissemination



DATA AGGREGATION AND DISSEMINATION BUSINESS PROCESS NOTES AND ANNOTATIONS

General notes: National digital case-based surveillance systems have several advantages compared with more traditional paper-based aggregated systems, such as reductions in the recording and reporting workload of frontline workers, improved data quality, faster access to data at all levels, more flexible data analysis, and enhanced use of data through record linkages between databases. WHO encourages countries to make the transition from paper-based aggregated to case-based digital surveillance. These activities can be taken at the community, health facility, or district/area/region health office level, or at the national level with the ministry of health or similar national organization.

1. Aggregate the data:

- The surveillance data manager aggregates the data for a specified period and surveillance population.

2. Check the data quality:

- Health-care facility data are reviewed for accuracy, validity and completeness.
- This can be supported through automated checks in a digital system.

3. Were any data quality issues found?

4. Correct fixable data quality issues:

- Where possible, inaccurate, invalid or incomplete data should be checked against source records and corrected according to the national standard operating procedures.

5. Deidentify the data and prepare a database:

- Deidentify and prepare the data for analysis and reporting.
- The database should include denominators needed for indicators, especially “total number of births (livebirths + stillbirths)” in the surveillance population during the time period for which the database and reports are being generated.

6. Generate aggregate reports:

- The surveillance data manager generates aggregate reports of predefined surveillance indicators aligned with national monitoring and evaluation guidelines.
- This can be automated and done digitally.

7. Review the quality of the report data:

- The surveillance programme manager reviews report quality and checks for any potential remaining data quality issues, such as implausible values or outputs.

8. Is the quality of the report data satisfactory?

- If so, the workflow continues with Step 9: approve the generated report.
- If not, the surveillance programme manager details the report issues before sending it back for correction, in Step 14: detail issues with the report.

9. Approve the generated report:

- The surveillance programme manager approves the report and adds a summary interpretation or comments as needed.

10. Disseminate the generated report:

- The report is disseminated as per the dissemination plan.

11. Analyse the data:

- Data analysis and interpretation of the reports should identify opportunities to improve the performance of the health-care facility, such as tracing missing data, and to monitor the prevalence of birth defects under surveillance.
- Data analysis and interpretation can be done regularly or on an ad hoc basis and should not be limited to the reporting schedule.

12. Did data analysis result in information that needs to be disseminated?

13. Disseminate the analysis:

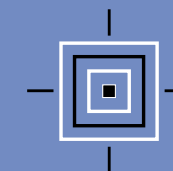
- Disseminate the analysis as per the dissemination plan or distribution list.
- This can be automated and done digitally.

14. Detail issues with the report:

- The issues identified in the report are detailed so that the surveillance data manager can apply the required corrections.
- The facility may be required to restart the process and resubmit the reports.

Additional considerations for adapting workflows

These workflows are meant to be generic and high-level. A degree of customization and adaptation will be needed as they are translated into a digital system for a specific context. The workflows are considered to be 80% complete – the other 20% will need to be done through a series of person-centred design methods and mechanisms to complete the workflows for implementation. For example, additional workflows may need to be drawn out, or there may be additional activities expected of a health worker in a facility. Some workflows are not included due to the high level of contextualization required, such as billing, dispensing (if separate from service provision), configuration (of facility-level specifics), and follow-up (which can be automated). Alternatively, there may be activities and tasks that a health worker would not be expected to do. Although these workflows can be considered a starting point, it is helpful to conduct further validation through interviews with key personas or by shadowing their work to obtain a better sense of the differences that need to be reflected in the digital system.



This section outlines the minimum set of data corresponding to different points of the workflow within the identified business processes. The dataset can be used on any software system and lists the data elements relevant for service delivery and executing decision-support logic, and for populating indicators and performance metrics. Although this section provides a high-level overview of the data elements, a more complete data dictionary in spreadsheet form detailing the input options, validation checks and concept dictionary codes is available here: <https://smart.who.int/dak-bds/dictionary.html>.

Inclusion of a data element in the table does not mean collection of the data is required. Additionally, some data elements are dependent on other data elements (e.g. test results are entered only when a test has been performed). Collection of data should not prevent clients from receiving treatment or affect clinical care.

Simplified list of core data elements

Table 8 provides a simplified list of core data elements and is a snapshot of the comprehensive data dictionary. As with the workflows, this data dictionary can be viewed as 80% generic, with the expectation that the other 20% will be supplemented and modified through country adaptation.

Table 8. Core data elements for identified business processes

Activity ID and name	Data element ID	Data element name	Description and definition
Business process BDS.A: data collection and abstraction			
BDS.A1. Receive and review the birth defect notification	N/A – no data are recorded during this activity		
BDS.A2. Verify whether a case record already exists	N/A – no data are recorded during this activity		

Activity ID and name	Data element ID	Data element name	Description and definition
BDS.A4. Create new case	BDS.A4.DE.1	Case record ID	Unique ID of the case moving through the health system. A case refers to a unique newborn regardless of the number of congenital anomalies they possess. This can be based on a national unique ID, a national health ID, biometrics, a system-generated unique ID or another value based on local context.
	BDS.A4.DE.2	Date of report	Date of reporting of the birth defect.
	BDS.A4.DE.4	Health facility ID	This is a context-specific unique identifier of the health facility at which surveillance is being conducted if the surveillance is facility-based.
BDS.A6. Review the case record	N/A – no data are recorded during this activity		
BDS.A7. Examine the newborn	N/A – no data are recorded during this activity		
BDS.A10. Seek consent for data collection	BDS.A10.DE.5	Consent provided for data collection	Indicates whether the informant gave their consent based on the information provided by the data collector if required by surveillance protocol.
	BDS.A10.DE.6	Signature for consent	A digital representation of the signature for consent for data collection.
BDS.A12. Collect outstanding data	BDS.A12.DE.7	First name	Newborn's first name.
	BDS.A12.DE.8	Last name	Newborn's family name or last name.
	BDS.A12.DE.9	Unique ID	Unique ID of the newborn moving through the health system. This is a context-specific data element and can be based on a national unique ID, a national health ID, biometrics, a system-generated unique ID or something else based on local policy.
	BDS.A12.DE.10	Date of birth	Newborn's date of birth, if known.
	BDS.A12.DE.13	Sex	Sex of newborn assigned at birth.
BDS.A15. Seek consent for photograph(s)	BDS.A15.DE.102	Consent provided for photographs	Indicates whether the informant gave their consent for photographs based on the information provided by the data collector if required by surveillance protocol.
	BDS.A15.DE.103	Signature for photographs	Digital representation of the signature for consent for photographs.
Business process BDS.B: case confirmation			
BDS.B1. Verify case in birth defects surveillance data	N/A – no data are recorded during this activity		
BDS.B4. Review the case record	N/A – no data are recorded during this activity		
BDS.B6. Update the case record	BDS.B6.DE.1	Diagnosis correction	This is a flag variable that is set true when the birth defects specialist changes any diagnosis. This can also be managed through analysis of the audit trails of the changes made to diagnosis information in the BDS.Common worksheet from the data dictionary.

Activity ID and name	Data element ID	Data element name	Description and definition
BDS.B11. Request further information	BDS.B11.DE.2	Request for further information to data manager	If the birth defects specialist requires further information to make a diagnosis, this field is used to describe what further information is needed. This field is the minimum requirement for this function. More sophisticated systems could add comments or questions to nearly all the fields in the BDS. Common worksheet from the data dictionary and the BDS.A workflow to indicate more precisely the issues with existing data and direct the data manager to make or solicit corrections.
BDS.B12. Prepare details for follow-up	BDS.B12.DE.3	Request for further information to data collector	If the birth defects data manager requires further information about the case from the data collector, this field is used to describe what further information is needed. This field is the minimum requirement for this function. More sophisticated systems could add comments or questions to nearly all the fields in the BDS. Common worksheet from the data dictionary and the BDS.A workflow to indicate more precisely the issues with existing data and direct the data collector to make corrections.
Business process BDS.C: data aggregation and dissemination			
BDS.C1. Aggregate the data	N/A – no data are recorded during this activity		
BDS.C2. Check the data quality	N/A – no data are recorded during this activity		
BDS.C7. Review the quality of the report data	BDS.C7.DE.1	Report satisfactory	Is the quality of the report data satisfactory?
BDS.C9. Approve the generated report	BDS.C9.DE.2	Signature	Approval of generated report
	BDS.C9.DE.3	Comments	Summary interpretation or comments
BDS.C10. Disseminate the generated report	N/A – no data are recorded during this activity		
BDS.C11. Analyse the data	N/A – no data are recorded during this activity		
BDS.C13. Disseminate the analysis	N/A – no data are recorded during this activity		
BDS.C14. Detail issues with the report	BDS.C14.DE.4	Request for corrections to data manager	If there are issues with the generated report, the surveillance programme manager documents them in this field to communicate to surveillance data manager.

List of calculated data elements

The previous section outlines the core data elements that should be included within digital systems to facilitate the decision-support logic or indicators. Table 9 shows the additional derived data elements based on calculations from core data elements.

Table 9. Calculated data elements

Calculated data element label	Core data elements used for calculation (variables)	Calculation
Father's age	Father's date of birth Date of birth	IF "Father's date of birth" IS NOT NULL AND "Date of birth" IS NOT NULL THEN ("Date of birth" – "Father's date of birth") / 365.25
Mother's age	Mother's date of birth Date of birth	IF "Mother's date of birth" IS NOT NULL AND "Date of birth" IS NOT NULL THEN ("Date of birth" – "Mother's date of birth") / 365.25

Additional considerations for adapting the data dictionary

Some settings may require the inclusion of additional data elements in the full dataset or changes to response options based on contextual differences. Additionally, the transition from paper-based forms to digital systems may require some reflection on whether data elements currently on paper forms should be incorporated into the digital system. If starting from paper-based registers and forms, additional guidance can be found in the WHO Digital transformation handbook for primary health care (7).

Table 10 lists the initial considerations anticipated for each implementation to review and customize based on national guidelines and local context.

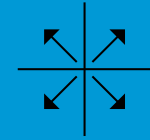
Table 10. Characteristics for local customization and configuration

Points of customization and configuration	Description
Unique identifier	The unique identifier of the client can be based on a national unique ID, a national health ID, biometrics, a system-generated unique identifier or something else.
National ID	The format of the national ID varies from country to country.
Facility identifier	The unique identifier of the facility. A reference to a facility registry or a reporting system should be included where possible.
Facility name	The name of the different health-care facilities based on a facility registry or a reporting system should be included where possible.
Ownership	This denotes whether the facility is public or private, where relevant.
Type of health-care facility	Type of facility, based on country terminology (e.g. health centre, health post, dispensary, hospital).
Global positioning system coordinates	Latitude and longitude coordinates can be included, if relevant for mapping purposes. This can be especially helpful in the context of community health workers who could be given birth defects surveillance tasks based on their catchment area and client visit history.
Administrative areas	Administrative areas can be based on geographical location, catchment area or another mechanism used by the country to manage health-care facilities.

Component

6

Decision-support logic

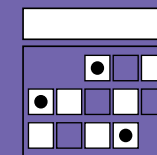


The decision-support logic component of the DAK provides the decision logics and algorithms in accordance with WHO guidelines.

The birth defects surveillance DAK does **not** include content for this component because it does not cover recommended clinical practice related to birth defects. The treatment of birth defects is complex and is unique to each case.

Component

7 Scheduling logic



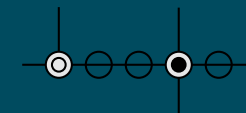
The scheduling logic component of the DAK provides information on scheduling of services, in accordance with WHO guidelines.

The birth defects surveillance DAK does **not** include content for this component because it does not cover recommended clinical practice related to birth defects. The necessity and recurrence of clinical follow-up care will vary, depending on the type and severity of the birth defect.

Component

8

Indicators and performance metrics

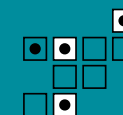


This section details indicators and performance metrics that would be aggregated from core data elements identified in Component 5: core data elements. Table 11 lists a minimum set of indicators that can be aggregated for decision-making, performance metrics, and subnational and national reporting based on data collected from individual-level routine health systems. These indicators may be aggregated automatically from the point-of-service system to populate a digital HMIS. The complete list of indicators and associated details is available in the indicators and performance metrics implementation tool here: <https://smart.who.int/dak-bds/indicators.html>.

Table 11. Indicators and performance metrics

Indicator ID	Indicator name	Definition	Numerator description	Denominator description	Disaggregation	References
BDS.IND.1	Prevalence of anencephaly	Prevalence of anencephaly per 10 000 births.	Number of confirmed and completed cases of anencephaly in surveillance population.	Total number of births in surveillance population.	• Outcome at birth • Congenital anomaly clinical presentation • Mother's age • Mother's race • Mother's ethnicity • Sex.	23,28,48
BDS.IND.2	Prevalence of craniorachischisis	Prevalence of craniorachischisis per 10 000 births.	Number of confirmed and completed cases of craniorachischisis in surveillance population.	Total number of births in surveillance population.	• Outcome at birth • Congenital anomaly clinical presentation • Mother's age • Mother's race • Mother's ethnicity • Sex.	23,28

Indicator ID	Indicator name	Definition	Numerator description	Denominator description	Disaggregation	References
BDS.IND.3	Prevalence of iniencephaly	Prevalence of iniencephaly per 10 000 births.	Number of confirmed and completed cases of iniencephaly in surveillance population.	Total number of births in surveillance population.	• Outcome at birth • Congenital anomaly clinical presentation • Mother's age • Mother's race • Mother's ethnicity • Sex.	23,28
BDS.IND.4	Prevalence of encephalocele	Prevalence of encephalocele per 10 000 births.	Number of confirmed and completed cases of encephalocele in surveillance population.	Total number of births in surveillance population.	• Outcome at birth • Congenital anomaly clinical presentation • Mother's age • Mother's race • Mother's ethnicity • Sex.	22,27
BDS.IND.5	Prevalence of spina bifida	Prevalence of spina bifida per 10 000 births.	Number of confirmed and completed cases of spina bifida in surveillance population.	Total number of births in surveillance population.	• Outcome at birth • Congenital anomaly clinical presentation • Mother's age • Mother's race • Mother's ethnicity • Sex.	23,28



Component

9

High-level functional and non-functional requirements

This section provides an overview of illustrative functional and non-functional requirements that may be considered to kickstart the process of designing or adapting the PCPOSS. Functional requirements describe the capabilities the system must have to meet end-user needs and achieve tasks within the business process. Non-functional requirements provide the general attributes and features of the digital system to ensure usability and overcome technical and physical constraints. Examples of non-functional requirements include ability to work offline, multiple language settings and password protection.

Table 12 highlights some of the key functional requirements for executing the business processes listed in [Component 4: generic business processes and workflows](#). Table 13 provides non-functional requirements as general characteristics of the overall system. The complete set of functional and non-functional requirements can be accessed in the functional and non-functional requirements implementation tool, available here: <https://smart.who.int/dak-bds/system-requirements.html>. These lists are not exhaustive and should be modified according to context and user persona needs.

Functional requirements

Table 12. Functional requirements

Requirement ID	Activity ID and name	As a ...	I want ...	So that ...
Business process BDS.A: data collection and abstraction				
BDS.FXNREQ.1	BDS.A2. Verify whether a case record already exists	Data collector	The system to require me (a user) to search to see whether the case record is already in the system before starting a new entry.	I can avoid duplicates.
BDS.FXNREQ.2	BDS.A2. Verify whether a case record already exists	Data collector	The system to return all potential matches based on search criteria.	I can find the best match.
BDS.FXNREQ.3	BDS.A2. Verify whether a case record already exists	Data collector	The system to display sufficient data to identify the case.	I can check whether this is a new or existing case record and avoid duplicates.
BDS.FXNREQ.4	BDS.A2. Verify whether a case record already exists	Data collector	The search to match on partial information (e.g. partial birth dates).	I have a better chance of finding a match.

Requirement ID	Activity ID and name	As a ...	I want ...	So that ...
BDS.FXNREQ.5	BDS.A2. Verify whether a case record already exists	Data collector	The system to allow search parameters configuration (e.g. mandatory fields, when partial information is acceptable).	A search can be optimized without compromising confidentiality.
Business process BDS.B: case confirmation				
BDS.FXNREQ.18	BDS.B4. Review the case record	Surveillance data manager	The birth defects surveillance system to notify the birth defects specialist that a case needs to be reviewed using an application notification, email, SMS or other method.	The birth defects specialist is informed in a timely manner about cases pending their review.
BDS.FXNREQ.19	BDS.B8. Review the birth defect(s) against case inclusion criteria	Surveillance data manager	The system to support with the necessary information related to case inclusion criteria, if it is stored in the system.	I can review birth defects against case inclusion criteria.
BDS.FXNREQ.20	BDS.B10. Provide updated information to the care provider	Surveillance data manager	To be able to send the final determination regarding the diagnosis back to the care provider using an application notification, email, SMS or other method, if required by the local protocol.	The care provider has a final verified diagnosis to provide better care to newborn.
Business process BDS.C: data aggregation and dissemination				
BDS.FXNREQ.24	BDS.C1. Aggregate data	Surveillance data manager	To have the ability to aggregate the data by time period and surveillance population.	I can check the data for quality issues.
BDS.FXNREQ.25	BDS.C1. Aggregate the data	Surveillance data manager	The system to automatically calculate indicators based on preset rules.	I do not have to calculate this myself.
BDS.FXNREQ.26	BDS.C2. Check the data quality	Surveillance data manager	To have the ability to query the data to look for issues of data quality.	I can detect problems with the data.

For the full list of functional requirements defined for the birth defects surveillance DAK, see the functional and non-functional requirements implementation tool.

Non-functional requirements

Table 13. Non-functional requirements

Requirement ID	Category	Non-functional requirement
BDS.NFXNREQ.1	Security – confidentiality	Provide password-protected access for authorized users.
BDS.NFXNREQ.2	Security – confidentiality	Provide a means to ensure confidentiality and privacy of personal health information.
BDS.NFXNREQ.3	Security – confidentiality	No personal data could be shared from the local device in the country to a cloud server or other servers out of the country.
BDS.NFXNREQ.10	Security – authentication	Notify the user to change their password the first time they log in.
BDS.NFXNREQ.11	Security – authentication	Adhere to complex password requirements.
BDS.NFXNREQ.12	Security – authentication	Provide a mechanism to securely change a user's password.
BDS.NFXNREQ.18	Security – audit trail and logs	Log system logins and logouts.
BDS.NFXNREQ.19	Security – audit trail and logs	Record all authentication violations.
BDS.NFXNREQ.20	Security – audit trail and logs	Log all activities performed by the user, including date- and timestamp.
BDS.NFXNREQ.27	Security – user management	Allow user with permission to create a new user and temporary password.
BDS.NFXNREQ.28	Security – user management	Support definitions of unlimited roles and assigned levels of access, viewing, entry, editing and auditing.

For the full set of non-functional requirements defined for the birth defects surveillance DAK, see the functional and non-functional requirements implementation tool.

References¹

1. Digital implementation investment guide (DIIG): Integrating digital interventions into health programmes. Geneva: World Health Organization; 2020 (<https://iris.who.int/handle/10665/334306>).
2. Collaborative Requirements Development Methodology (CRDM). Decatur, GA: Public Health Informatics Institute; 2016 (<https://www.phii.org/crdm/>).
3. Classification of digital interventions, services and applications in health: a shared language to describe the uses of digital technology for health, 2nd edition. Geneva: World Health Organization; 2023 (<https://iris.who.int/handle/10665/373581>).
4. WHO guideline: recommendations on digital interventions for health system strengthening. Geneva: World Health Organization; 2019 (<https://iris.who.int/handle/10665/311941>).
5. Digital health and interoperability: key terms and theory of change small working group. Health Data Collaborative Digital; 2019 (https://docs.google.com/presentation/d/1TnTFaunk-1WLIg4sKJQ_aSfjmfivvcENil4mY4XxJs).
6. Developing health management information systems: a practical guide for developing countries. Manila: World Health Organization Regional Office for the Western Pacific; 2004 (<https://iris.who.int/handle/10665/207050>).
7. Digital transformation handbook for primary health care: optimizing person-centred point of service systems. Geneva: World Health Organization; 2024 (<https://iris.who.int/handle/10665/379452>).
8. International statistical classification of diseases and related health problems (ICD). Geneva: World Health Organization; 2023 (<https://www.who.int/standards/classifications/classification-of-diseases>).
9. Seventy-first World Health Assembly. Resolution WHA71.7: digital health. New York: United Nations; 2018 (https://apps.who.int/gb/ebwha/pdf_files/WHA71/A71_R7-en.pdf).
10. Global strategy on digital health 2020–2025. Geneva: World Health Organization; 2021 (<https://iris.who.int/handle/10665/344249>).
11. World Bank Group, United States Agency for International Development, World Health Organization. The roadmap for health measurement and accountability: MA4Health. Washington, DC: World Bank; 2015 (<https://www.healthdatacollaborative.org/fileadmin/uploads/hdc/Documents/the-roadmap-for-health-measurement-andaccountability.pdf>).
12. Digital strategy 2020–2024. Washington, DC: United States Agency for International Development (https://www.usaid.gov/sites/default/files/2022-05/USAID_Digital_Strategy.pdf.pdf).
13. The Global Fund strategic framework for data use for action and improvement at country level 2017–2022. Geneva: Global Fund to Fight AIDS, Tuberculosis and Malaria, 2019 (https://www.theglobalfund.org/media/8362/me_datauseforactionandimprovement_framework_en.pdf).
14. Michaels M. Adapting clinical guidelines for the digital age: summary of a holistic and multidisciplinary approach. Am J Med Qual. 2023;38(5S Suppl. 2):S3–S11 ([doi:10.1097/jmq.000000000000138](https://doi.org/10.1097/jmq.000000000000138)).
15. Tan R, Cobuccio L, Beynon F, Levine G, Vaezipour N, Luwanda L, et al. ePOCT+ and the medAL-suite: development of an electronic clinical decision support algorithm and digital platform for pediatric outpatients in low- and middle-income countries. PLOS Digit Health. 2023;2(1):e0000170 ([doi:10.1371/journal.pdig.0000170](https://doi.org/10.1371/journal.pdig.0000170)).
16. Boxwala AA, Rocha BH, Maviglia S, Kashyap V, Meltzer S, Kim J, et al. A multi-layered framework for disseminating knowledge for computer-based decision support. J Am Med Inform Assoc. 2011;18(Suppl. 1):i132–i139 ([doi:10.1136/amiajnl-2011-000334](https://doi.org/10.1136/amiajnl-2011-000334)).

¹ All references were accessed on 15 January 2025.

17. Fenella B, Frédérique G, Riccardo L, Torsten S, Rainer T, Natschja R, et al. Digitalizing clinical guidelines: experiences in the development of clinical decision support algorithms for management of childhood illness in resource-constrained settings. *Glob Health Sci Pract*. 2023;11(4):e2200439 ([doi:10.9745/GHSP-D-22-00439](https://doi.org/10.9745/GHSP-D-22-00439)).
18. Mehl GL, Seneviratne MG, Berg ML, Bidani S, Distler RL, Gorgens M, et al. A full-STAC remedy for global digital health transformation: open standards, technologies, architectures and content. *Oxford Open Digital Health*. 2023;1:oqad018 ([doi:10.1093/oodh/oqad018](https://doi.org/10.1093/oodh/oqad018)).
19. Tamrat T, Ratanaprayul N, Barreix M, Tunçalp Ö, Lowrance D, Thompson J, et al. Transitioning to digital systems: the role of World Health Organization's digital adaptation kits in operationalizing recommendations and interoperability standards. *Glob Health Sci Pract*. 2022;10(1):e2100320 ([doi:10.9745/GHSP-D-21-00320](https://doi.org/10.9745/GHSP-D-21-00320)).
20. Barton C, Kallem C, Van Dyke P, Mon D, Richesson R. Demonstrating “collect once, use many”: assimilating public health secondary data use requirements into an existing domain analysis model. *AMIA Annu Symp Proc*. 2011:98–107.
21. Mehl G, Tunçalp Ö, Ratanaprayul N, Tamrat T, Barreix M, Lowrance D, et al. WHO SMART guidelines: optimising country-level use of guideline recommendations in the digital age. *Lancet Digit Health*. 2021;3(4):e213–e216 ([doi:10.1016/s2589-7500\(21\)00038-8](https://doi.org/10.1016/s2589-7500(21)00038-8)).
22. SMART guidelines [website]. Geneva: World Health Organization (<https://www.who.int/teams/digital-health-and-innovation/smart-guidelines>).
23. World Health Organization, United States Centers for Disease Control and Prevention, International Clearinghouse for Birth Defects Surveillance and Monitoring. Birth defects surveillance: a manual for programme managers, 2nd edition. Geneva: World Health Organization; 2020 (<https://iris.who.int/handle/10665/337425>).
24. UHC compendium: repository of interventions for universal health coverage [website]. Geneva: World Health Organization (<https://www.who.int/universal-health-coverage/compendium/interventions-by-programme-area>).
25. WHO UHC Service Planning, Delivery & Implementation (SPDI) Platform [website]. Geneva: World Health Organization (<https://uhcc.who.int/uhcpackages/>).
26. Sixty-third World Health Assembly. Geneva: World Health Organization; 2010 (https://apps.who.int/gb/e/e_wha63.html).
27. Accelerate progress towards reducing maternal, newborn and child mortality in order to achieve Sustainable Development Goal targets 3.1 and 3.2. Geneva: World Health Organization; 2024 (<https://www.who.int/publications/m/item/wha-agenda-item-11.7---accelerate-progress-towards-reducing-maternal-newborn-and-child-mortality-in-order-to-achieve-sdg-targets-3.1-and-3.2>).
28. World Health Organization, United States Centers for Disease Control and Prevention, International Clearinghouse for Birth Defects Monitoring and Surveillance. Birth defects surveillance: quick reference handbook of selected congenital anomalies and infections. Geneva: World Health Organization; 2020 (<https://iris.who.int/handle/10665/338485>).
29. Birth defects surveillance training: facilitator's guide, 2nd edition. Geneva: World Health Organization; 2024 (<https://www.who.int/publications/i/item/9789240099913>).
30. Dolk H, Leke AZ, Whitfield P, Moore R, Karnell K, Barišić I, et al. Global birth defects app: an innovative tool for describing and coding congenital anomalies at birth in low resource settings. *Birth Defects Res*. 2021;113(14):1057–1073 ([doi:10.1002/bdr2.1898](https://doi.org/10.1002/bdr2.1898)).
31. Global Birth Defects Description and Coding (GBDDC). Belfast: Ulster University; 2020 (<https://globalbirthdefects.tghn.org/download-birth-defects-surveillance-app/launch-basic-version-app/>).
32. Global Birth Defect Description and Coding (GBDDC). Belfast: Ulster University; 2020 Apple App Store (<https://apps.apple.com/us/app/birth-defect-description/id1445679235>).
33. World Health Organization, International Telecommunication Union. National eHealth strategy toolkit. Geneva: International Telecommunication Union; 2012 (<https://iris.who.int/handle/10665/75211>).
34. Analysis and use of health facility data: guidance for maternal, newborn, child and adolescent health programme managers. Geneva: World Health Organization; 2023 (<https://www.who.int/publications/i/item/9789240080331>).
35. SCORE for health data technical package. Geneva: World Health Organization; 2020 (<https://www.who.int/data/data-collection-tools/score>).
36. Stepwise toolkit for planning and budgeting interoperability of digital health solutions. Geneva: Digital Health Centre of Excellence (https://a78da35e-056b-4420-8e4f-41e3283327e3.usrfiles.com/ugd/55ae33_5af35824932c48d89287928244b60a8d.pdf).

37. SMART Guidelines Starter Kit [website]. Geneva: World Health Organization; 2024 (<https://smart.who.int/ig-starter-kit/>).
38. Monitoring and evaluating digital health interventions: a practical guide to conducting research and assessment. Geneva: World Health Organization; 2016 (<https://iris.who.int/handle/10665/252183>).
39. Congenital disorders. Geneva: World Health Organization; 2023 (<https://www.who.int/news-room/fact-sheets/detail/birth-defects>).
40. WHO recommendations: optimizing health worker roles to improve access to key maternal and newborn health interventions through task shifting. Geneva: World Health Organization; 2012 (<https://iris.who.int/handle/10665/77764>).
41. World Health Organization, Johns Hopkins Bloomberg School of Public Health Center for Communication Programs. Family planning: a global handbook for providers – evidence-based guidance developed through worldwide collaboration, 3rd edition. Geneva: World Health Organization; 2018 (<https://iris.who.int/handle/10665/260156>).
42. Annual report 2014. Rome: International Clearinghouse for Birth Defects Surveillance and Research; 2014 (<http://www.icbdsr.org/resources/annual-report/>).
43. Birth defects. Atlanta, GA: United States Centers for Disease Control and Prevention (https://www.cdc.gov/birth-defects/?CDC_AAref_Val=https://www.cdc.gov/ncbddd/birthdefects/surveillancemanual/index.html).
44. Guidelines for data registration. Strasbourg: European Network of Population-based Registries for the Epidemiological Surveillance of Congenital Anomalies (https://eu-rd-platform.jrc.ec.europa.eu/eurocat/data-collection/guidelines-for-data-registration_en#inline-nav-2).
45. Classifying health workers: mapping occupations to the international standard classification. Geneva: World Health Organization; 2019 (<https://cdn.who.int/media/docs/default-source/health-workforce/dek/classifying-health-workers.pdf>).
46. International Standard Classification of Occupations (ISCO). Geneva: International Labour Organization (<https://ilostat.ilo.org/methods/concepts-and-definitions/classification-occupation/>).
47. Business Process Model and Notation [website]. Milford, MA: Object Management Group (<https://www.omg.org/bpmn/>).
48. Diaz T, Strong KL, Cao B, Guthold R, Moran AC, Moller A-B, et al. A call for standardised age-disaggregated health data. *Lancet Health Long*. 2021;2(7):e436–e4e43



World Health Organization
Avenue Appia 20
1211 Geneva
Switzerland

info@who.int
digitalhealth@who.int
www.who.int